```
<app-demo></app-demo>
```

```
@Component({
    selector: 'app-demo',
    template: '<p>This is a paragraph</p>'
})
export class DemoComponent{
    constructor(){
        //Some code
    }

    title: string = 'App Demo';
    show: boolean = false;
    @input() myVar: string;
}
```

When a constructor is called, by that time, none of its input properties are updated and available to use.

When a constructor is called, by that time, the child components of that component are not yet constructed.

Projected contents are also not available by the time the constructor of a component is called.

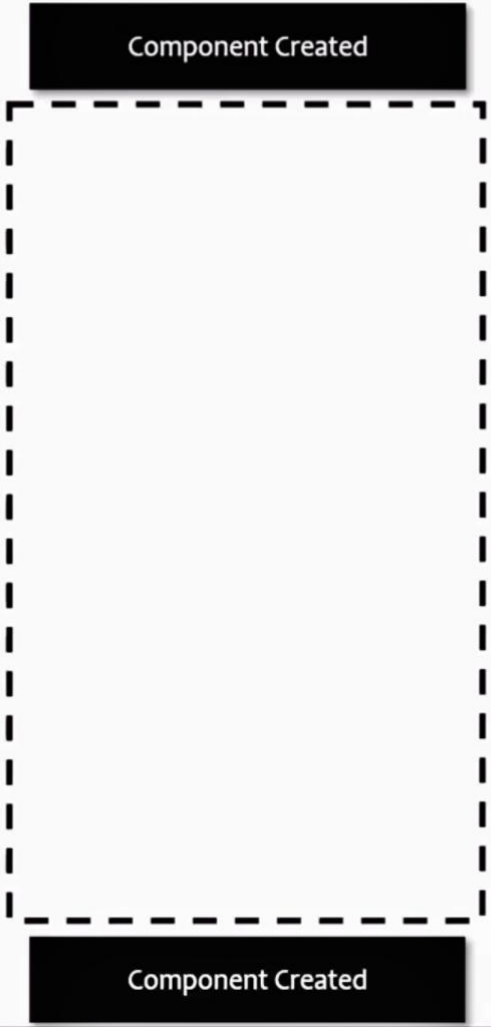
Once the component is removed from the DOM, we can say component is destroyed.

```
<app-demo>
    <p>This content will be projected.<p>
</app-demo>
```

```
@Component({
    selector: 'app-demo',
    template: '<h2>Content Projection</h2>
                <ng-content></ng-content>
                <p>This is a paragraph</p>'

})
export class DemoComponent{
    constructor(){
        //Some code
    }

    title: string = 'App Demo';
    show: boolean = false;
    @input() myVar: string;
}
```



- When the Angular application starts, it first creates and renders the root component.
- Then it creates and renders its children and their children. In this way, it forms a tree of components.
- Once Angular loads the component, it starts the process of rendering the view. To do that, it needs to check the input properties, evaluate the data bindings & expressions, render the projected content etc.
- Angular then also removes the component from the DOM when it is no longer needed.
- And Angular lets us know when these events happen, using Angular lifecycle hooks.

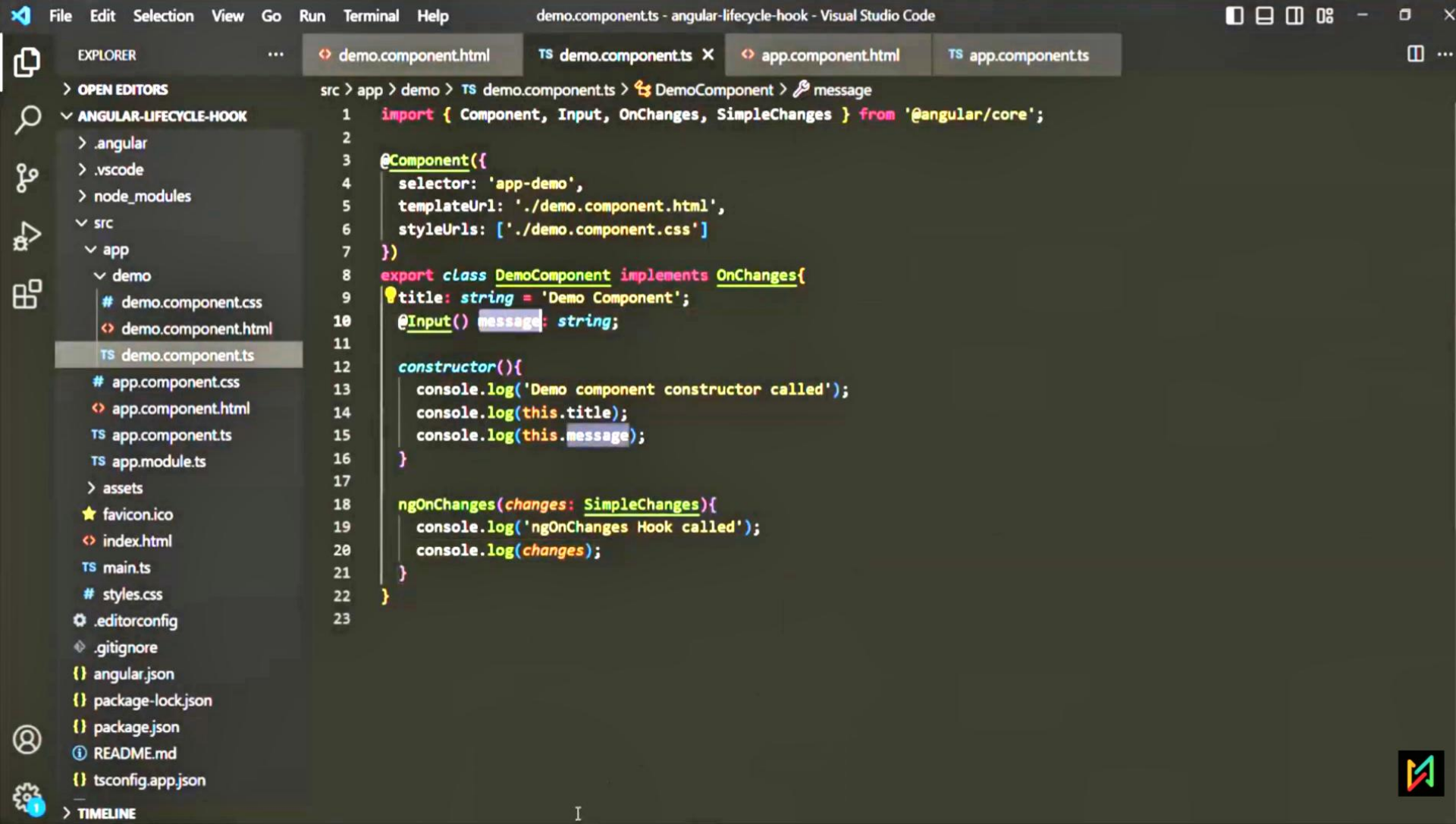
The Angular life cycle hooks are the methods that angular invokes on a directive or a component, as it creates, changes and destroys them.

The Angular life cycle hooks are the methods that angular invokes on a directive or a component, as it creates, changes and destroys them.

Change detection in Angular is a mechanism by which, Angular keeps the view template in sync with the component class.

```
<div>Hello {{ name }}</div>
```

- Whenever the @input property of a component changes
- Whenever a DOM event happens. E.g. Click or Change
- Whenever a timer events happens using `setTimeOut()` / `setInterval()`.
- Whenever an HTTP request is made.


File Edit Selection View Go Run Terminal Help
demo.component.ts - angular-lifecycle-hook - Visual Studio Code
EXPLORER
demo.component.html
demo.component.ts
app.component.html
app.component.ts
src > app > demo > demo.component.ts > DemoComponent > message
OPEN EDITORS
ANGULAR-LIFECYCLE-HOOK
.angular
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node_modules
src
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demo
demo.component.css
demo.component.html
demo.component.ts
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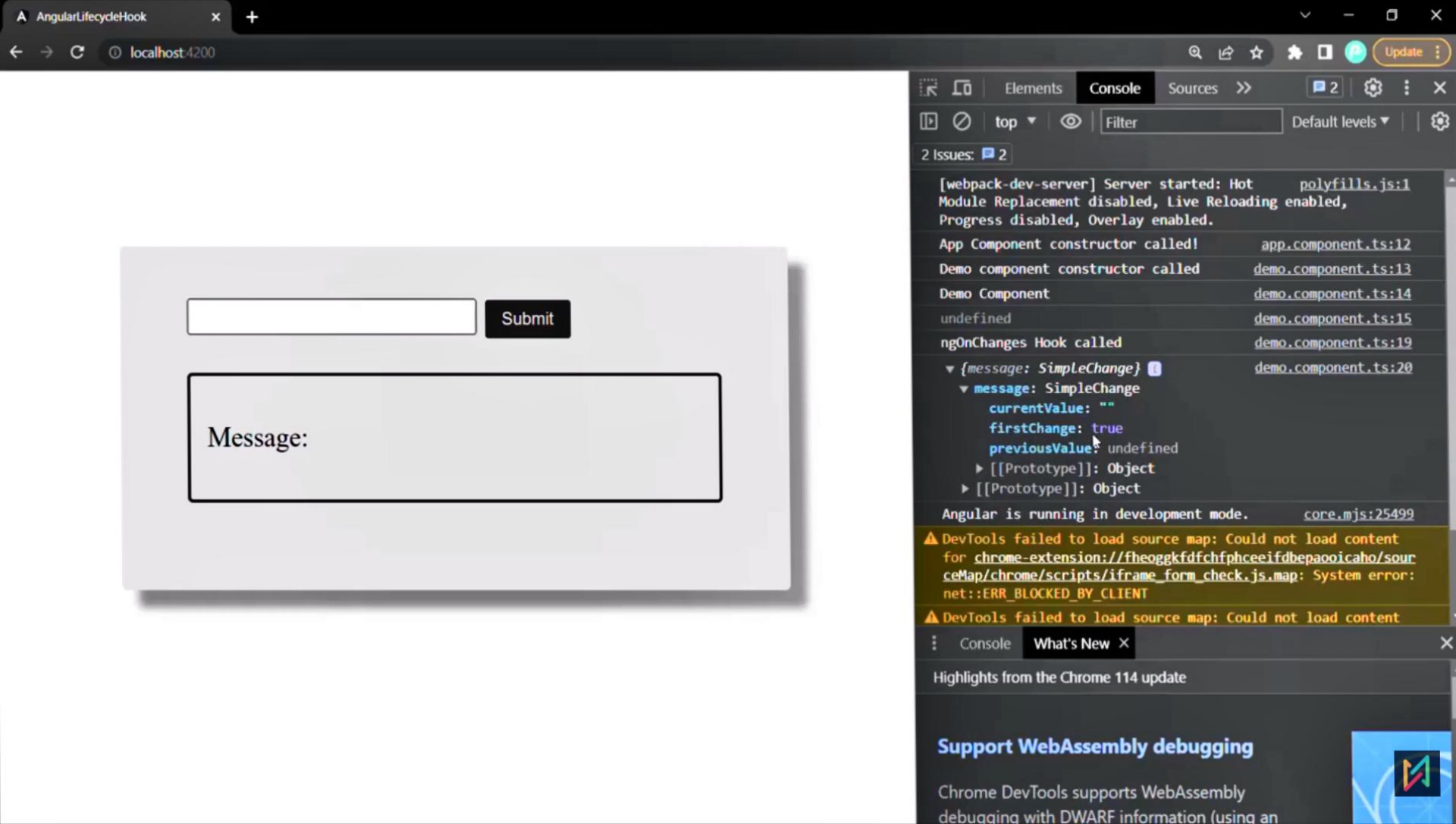
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
index.html
main.ts
styles.css
.editorconfig
.gitignore
angular.json
package-lock.json
package.json
README.md
tsconfig.app.json
TIMELINE
1 import { Component, Input, OnChanges, SimpleChanges } from '@angular/core';
2
3 @Component({
4 selector: 'app-demo',
5 templateUrl: './demo.component.html',
6 styleUrls: ['./demo.component.css']
7 })
8 export class DemoComponent implements OnChanges{
9 title: string = 'Demo Component';
10 @Input() message: string;
11
12 constructor(){
13 console.log('Demo component constructor called');
14 console.log(this.title);
15 console.log(this.message);
16 }
17
18 ngOnChanges(changes: SimpleChanges){
19 console.log('ngOnChanges Hook called');
20 console.log(changes);
21 }
22 }
23


AngularLifecycleHook
localhost:4200
Update
Submit
Message:
Elements
Console
Sources
2
top
Filter
Default levels
2 Issues: 2
[webpack-dev-server] Server started: Hot Module Replacement disabled, Live Reloading enabled, Progress disabled, Overlay enabled.
polyfills.js:1
App Component constructor called!
app.component.ts:12
Demo component constructor called
demo.component.ts:13
Demo Component
demo.component.ts:14
undefined
demo.component.ts:15
ngOnChanges Hook called
demo.component.ts:19
{message: SimpleChange}
demo.component.ts:20
message: SimpleChange
currentValue: ""
firstChange: true
previousValue: undefined
[[Prototype]]: Object
[[Prototype]]: Object
Angular is running in development mode.
core.mjs:25499
DevTools failed to load source map: Could not load content for chrome-extension://fheoggkfdfchfphceeifdbepaooicaho/sourceMap/chrome/scripts/iframe_form_check.js.map: System error: net::ERR_BLOCKED_BY_CLIENT
DevTools failed to load source map: Could not load content
Console
What's New
Highlights from the Chrome 114 update
Support WebAssembly debugging
Chrome DevTools supports WebAssembly debugging with DWARF information (using an

The ngOnChanges hook get executed at the start, when a new component is created and its input bound properties are updated.

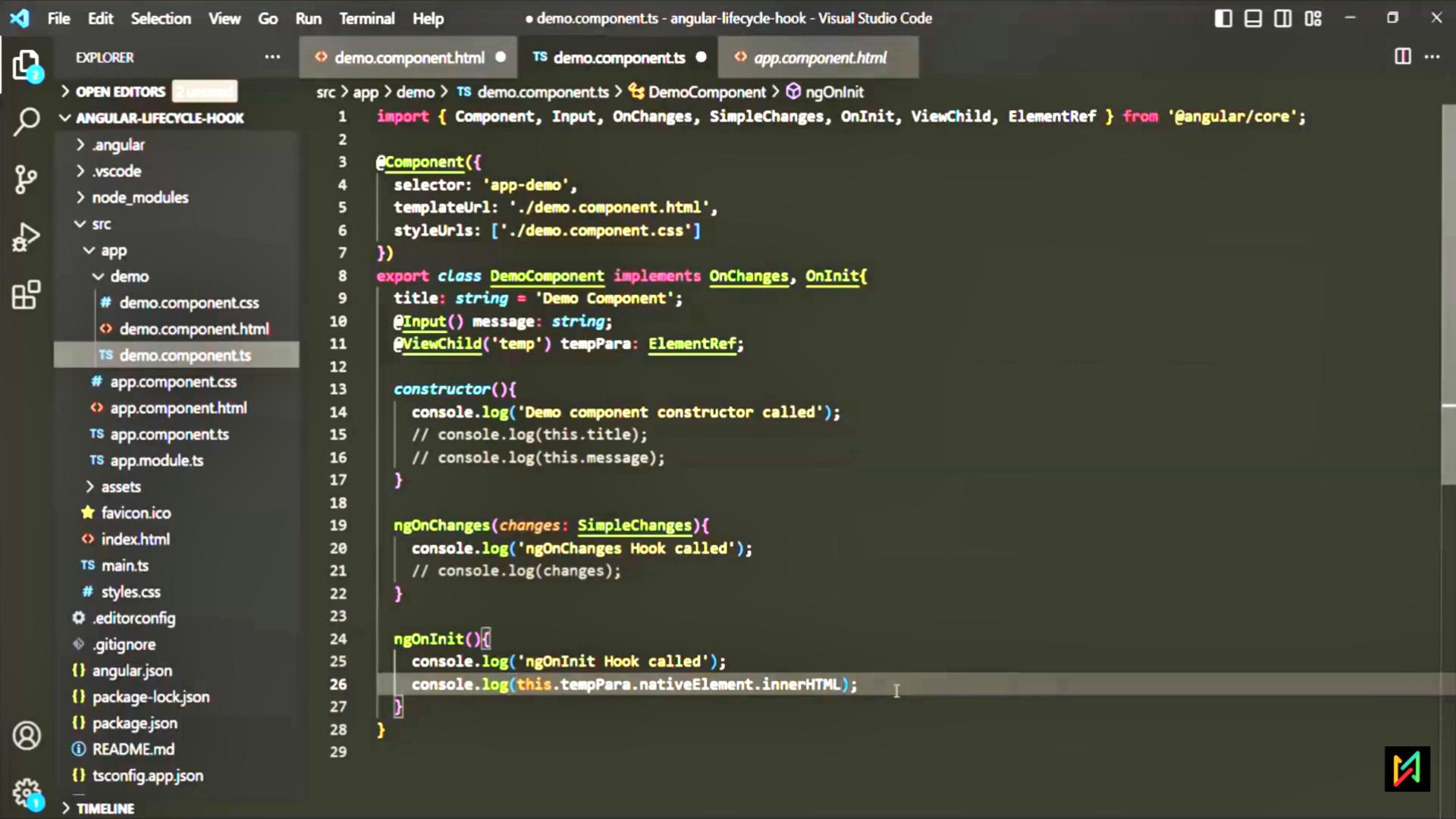
The ngOnChanges hook also gets executes everytime the input bound properties of the component changes

The ngOnChanges hook is not raised if the change detection cycle does not find any changes in the input propertie's value.

Angular raises ngOnInit hook after it creates the component and update its input properties. This hook is raised after ngOnChanges.

The ngOnInit hook is fired only once i.e. during the first change detection cycle. After that, if the input propery changes, this hook does not gets called.

By the time ngOnInit gets called, none of the child components or projected contents or view are available at this point. Hence any property decorated with @ViewChild, @ViewChildren, @ContentChild or @ContentChildren will not be available to use.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• demo.component.ts - angular-lifecycle-hook - Visual Studio Code
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app.component.ts
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assets
favicon.ico
index.html
main.ts
styles.css
.editorconfig
.gitignore
angular.json
package-lock.json
package.json
README.md
tsconfig.app.json
TIMELINE
src > app > demo > demo.component.ts > DemoComponent > ngOnInit
1 import { Component, Input, OnChanges, SimpleChanges, OnInit, ViewChild, ElementRef } from '@angular/core';
2
3 @Component({
4 selector: 'app-demo',
5 templateUrl: './demo.component.html',
6 styleUrls: ['./demo.component.css']
7 })
8 export class DemoComponent implements OnChanges, OnInit{
9 title: string = 'Demo Component';
10 @Input() message: string;
11 @ViewChild('temp') tempPara: ElementRef;
12
13 constructor(){
14 console.log('Demo component constructor called');
15 // console.log(this.title);
16 // console.log(this.message);
17 }
18
19 ngOnChanges(changes: SimpleChanges){
20 console.log('ngOnChanges Hook called');
21 // console.log(changes);
22 }
23
24 ngOnInit(){
25 console.log('ngOnInit Hook called');
26 console.log(this.tempPara.nativeElement.innerHTML);
27 }
28 }
29

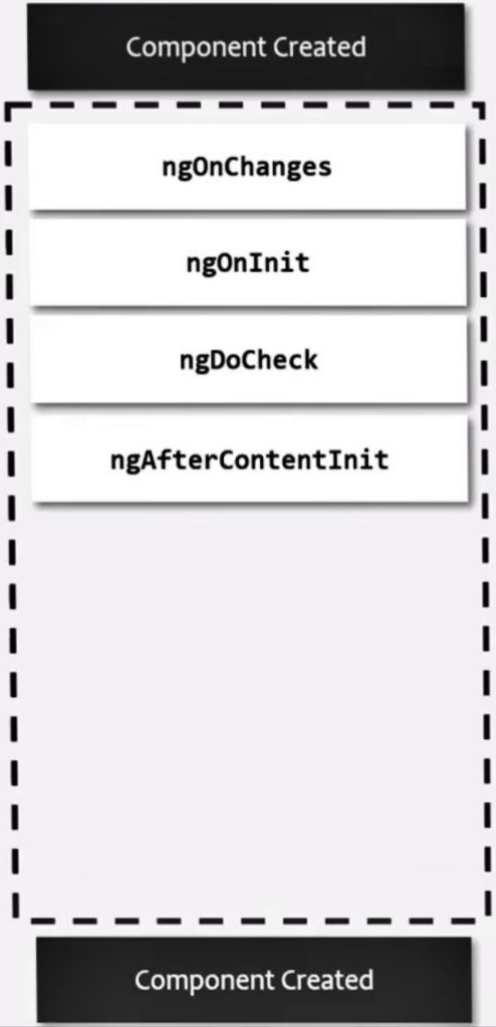
Angular invokes **ngDoCheck** hook during every change detection cycle. This hook is invoked even if there is no change in input bound properties.

For example: The **ngDoCheck** lifecycle hook will run if you clicked a button on webpage, which does not do anything. But still its an event so the change detection cycle will run and execute **ngDoCheck** hook.

Angular invokes **ngDoCheck** lifecycle hook after **ngOnChanges** & **ngOnInit** hooks.

You can use this hook to implement a custom change detection, whenever angular fails to detect any change made to input bound properties.

The ngDoCheck hook is also a great place to use, when you want to execute some code on every change detection cycle.



The ngAfterContentInit lifecycle hook is called after the components projected content has been fully initialized.

parent.component.html

```
<h2>Parent Component</h2>
<app-child>
    <h2>Some Heading</h2>
    <p #paragraph>This is a paragraph</p>
    <app-test></app-test>
</app-child>
```

child.component.html

```
<h2>Parent Component</h2>
<ng-content></ng-content>
```

The ngAfterContentInit lifecycle hook is called after the components projected content has been fully initialized.

Angular updates the properties decorated with @ContentChild & @ContentChildren decorator just before this hook is raised.

This lifecycle hook gets called only once, during the first change detection cycle. After that, if the projected content changes, this lifecycle hook will not get called.

The **ngAfterContentChecked** lifecycle hook is called during every change detection cycle, after Angular has finished initializing and checking projected content.

Angular also updates the properties decorated with **@ContentChild** & **@ContentChildren** decorator, before raising **ngAfterContentChecked** hook.

Angular raises this hook even if there is no projected content in the component.

The **ngAfterContentInit** hook is called after the projected content is initialized. **ngAfterContentChecked** is called whenever the projected content is *initialized*, *checked* & *updated*.

NOTE: The **ngAfterContentInit** & **ngAfterContentChecked** are component only hook. These hooks are not available for directives.

The **ngAfterViewInit** is called after the components View template and all its child components view templates are fully initialized.

Angular also updates the properties decorated with **@ViewChild** and **@ViewChildren** decorator before raising this hook.

This hook is called during the first change detection cycle, when Angular initializes the view for the first time.

By the time this hook get's called for a component, all the lifecycle hook methods of child components and directives are completely processed and child components are completely ready.

Angular fires **ngAfterViewChecked** hook after it checks and updates the components View template and all its child components view templates.

This hook is called during the first change detection cycle, after **ngAfterViewInit** hook has executed. And after that during every change detection cycle.

Angular also updates the properties decorated with **@ViewChild** and **@ViewChildren** decorator before raising this hook.

NOTE: The **ngAfterViewChecked** hook is a component only hook. It is not available for directives.

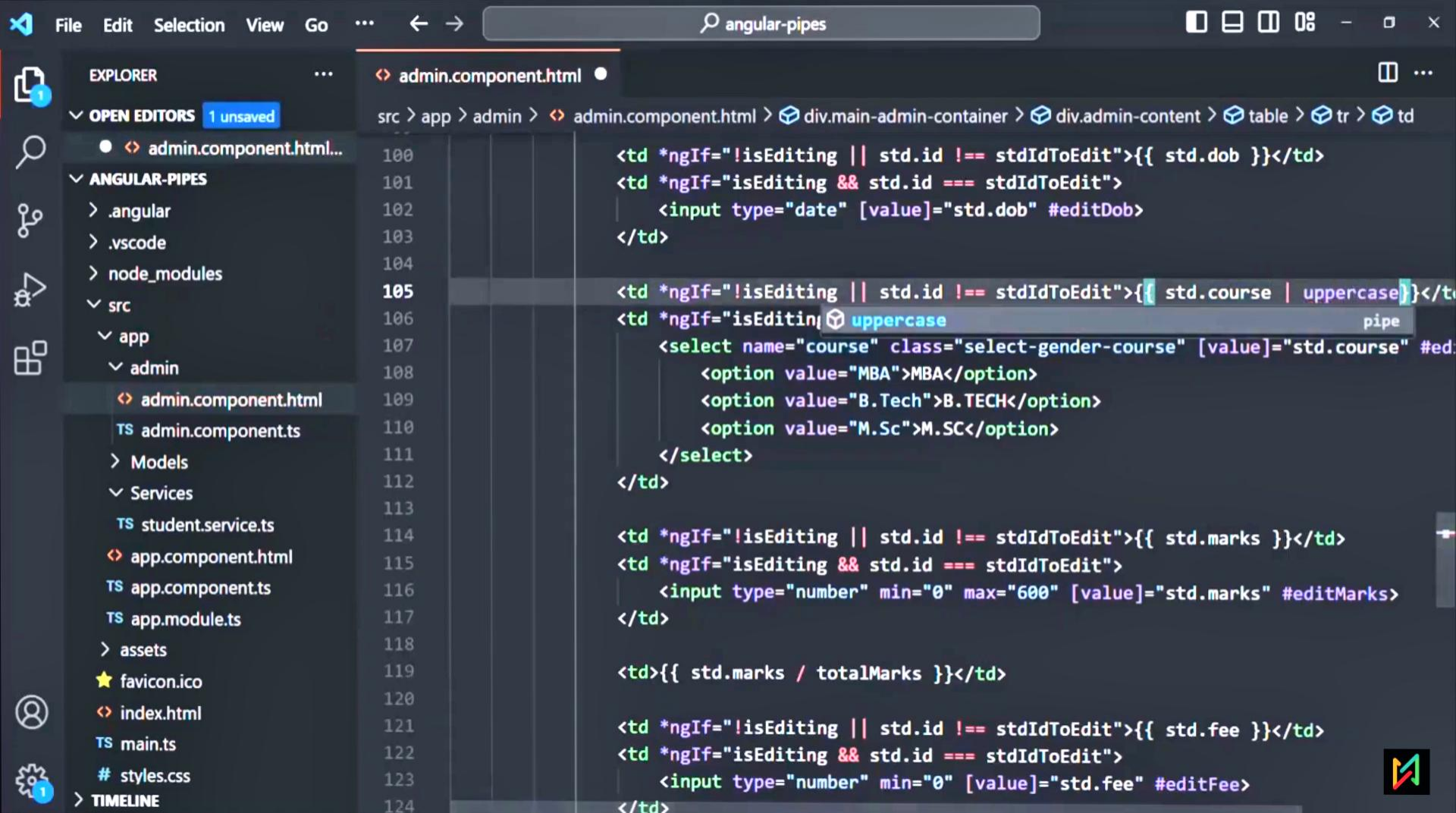
Angular fires **ngOnDestroy** lifecycle hook just before the component or the directive is destroyed i.e. removed from the DOM.

This hook is a great place to do some cleanup work like unsubscribe from an observable or detach event handler etc., as this hook is called right before the component is destroyed.

This **ngOnDestroy** is the last lifecycle hook of a component & a directive.

What is a pipe?

Pipes in Angular are used to transform or format data before displaying it in the view. Angular pipe takes data as an input and it formats or transforms that data before displaying it in the template.


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
OPEN EDITORS 1 unsaved
admin.component.html...
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
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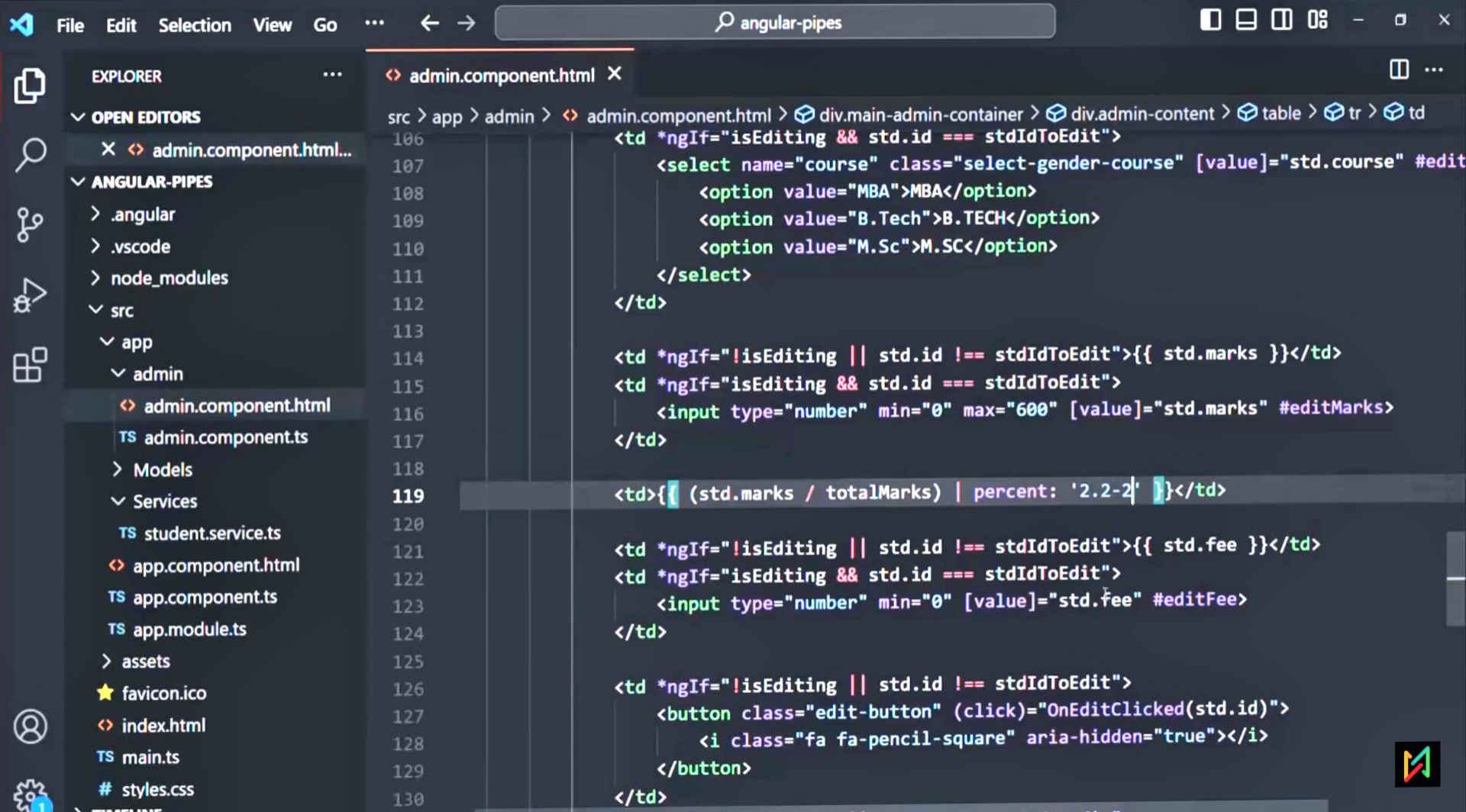
Models
Services
student.service.ts
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
index.html
main.ts
styles.css
TIMELINE
src > app > admin > admin.component.html > div.main-admin-container > div.admin-content > table > tr > td
<td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.dob }}</td>
<td *ngIf="isEditing && std.id === stdIdToEdit">
<input type="date" [value]="std.dob" #editDob>
</td>
<td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.course | uppercase}}</t
<td *ngIf="isEditin
uppercase
pipe
<select name="course" class="select-gender-course" [value]="std.course" #ed
<option value="MBA">MBA</option>
<option value="B.Tech">B.TECH</option>
<option value="M.Sc">M.SC</option>
</select>
</td>
<td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.marks }}</td>
<td *ngIf="isEditing && std.id === stdIdToEdit">
<input type="number" min="0" max="600" [value]="std.marks" #editMarks>
</td>
<td>{{ std.marks / totalMarks }}</td>
<td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.fee }}</td>
<td *ngIf="isEditing && std.id === stdIdToEdit">
<input type="number" min="0" [value]="std.fee" #editFee>
</td>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angular-pipes
EXPLORER
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ANGULAR-PIPES
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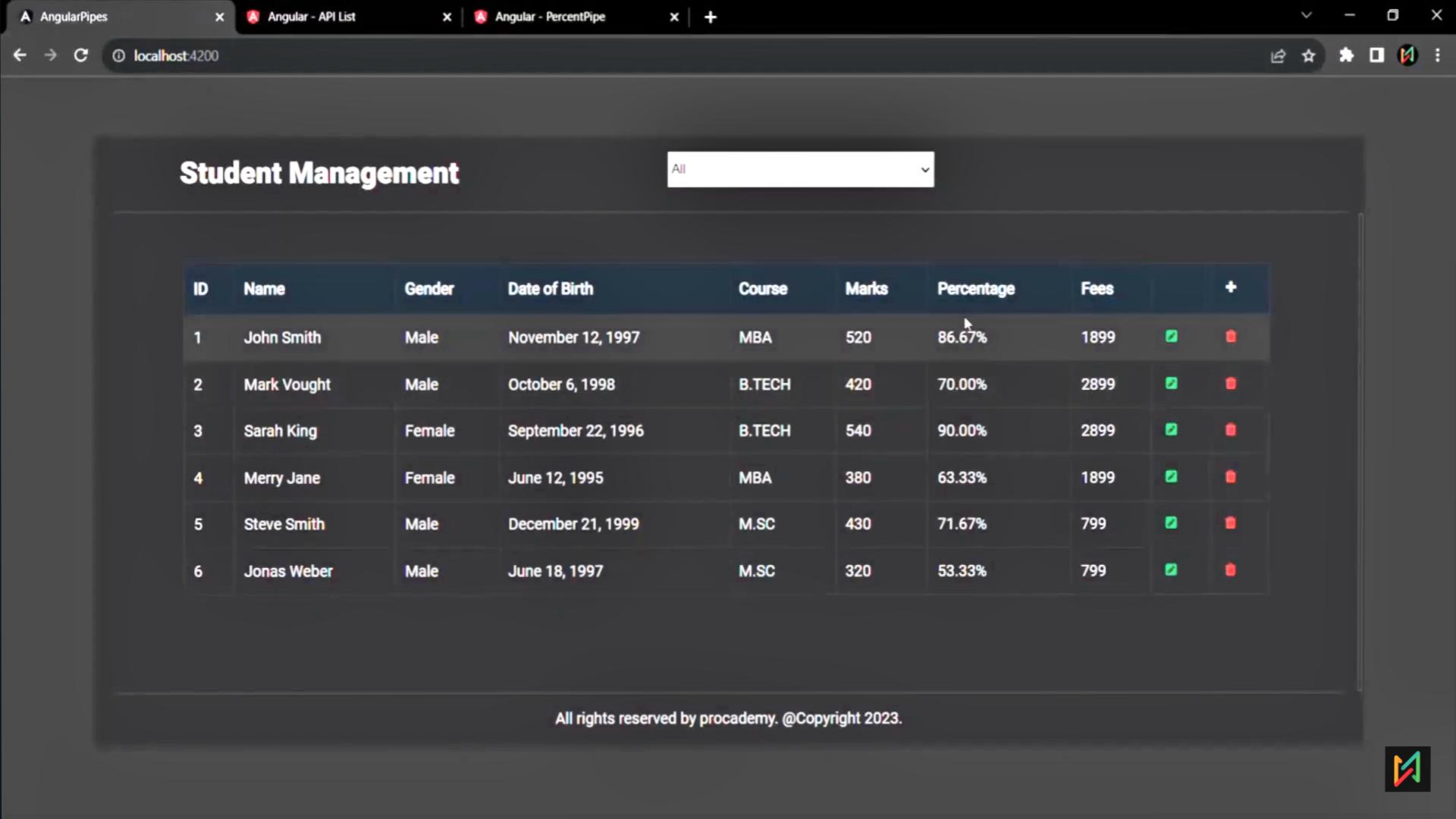
src > app > admin > admin.component.html > div.main-admin-container > div.admin-content > table > tr > td
<td *ngIf="isEditing && std.id === stdIdToEdit">
<select name="course" class="select-gender-course" [value]="std.course" #edit
<option value="MBA">MBA</option>
<option value="B.Tech">B.TECH</option>
<option value="M.Sc">M.SC</option>
</select>
</td>
<td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.marks }}</td>
<td *ngIf="isEditing && std.id === stdIdToEdit">
<input type="number" min="0" max="600" [value]="std.marks" #editMarks>
</td>
<td>{{ (std.marks / totalMarks) | percent: '2.2-2' }}</td>
<td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.fee }}</td>
<td *ngIf="isEditing && std.id === stdIdToEdit">
<input type="number" min="0" [value]="std.fee" #editFee>
</td>
<td *ngIf="!isEditing || std.id !== stdIdToEdit">
<button class="edit-button" (click)="OnEditClicked(std.id)">

</button>
</td>


AngularPipes
Angular - API List
Angular - PercentPipe
localhost:4200
Student Management
All
ID
Name
Gender
Date of Birth
Course
Marks
Percentage
Fees
1 John Smith Male November 12, 1997 MBA 520 86.67% 1899
2 Mark Vought Male October 6, 1998 B.TECH 420 70.00% 2899
3 Sarah King Female September 22, 1996 B.TECH 540 90.00% 2899
4 Merry Jane Female June 12, 1995 MBA 380 63.33% 1899
5 Steve Smith Male December 21, 1999 M.SC 430 71.67% 799
6 Jonas Weber Male June 18, 1997 M.SC 320 53.33% 799
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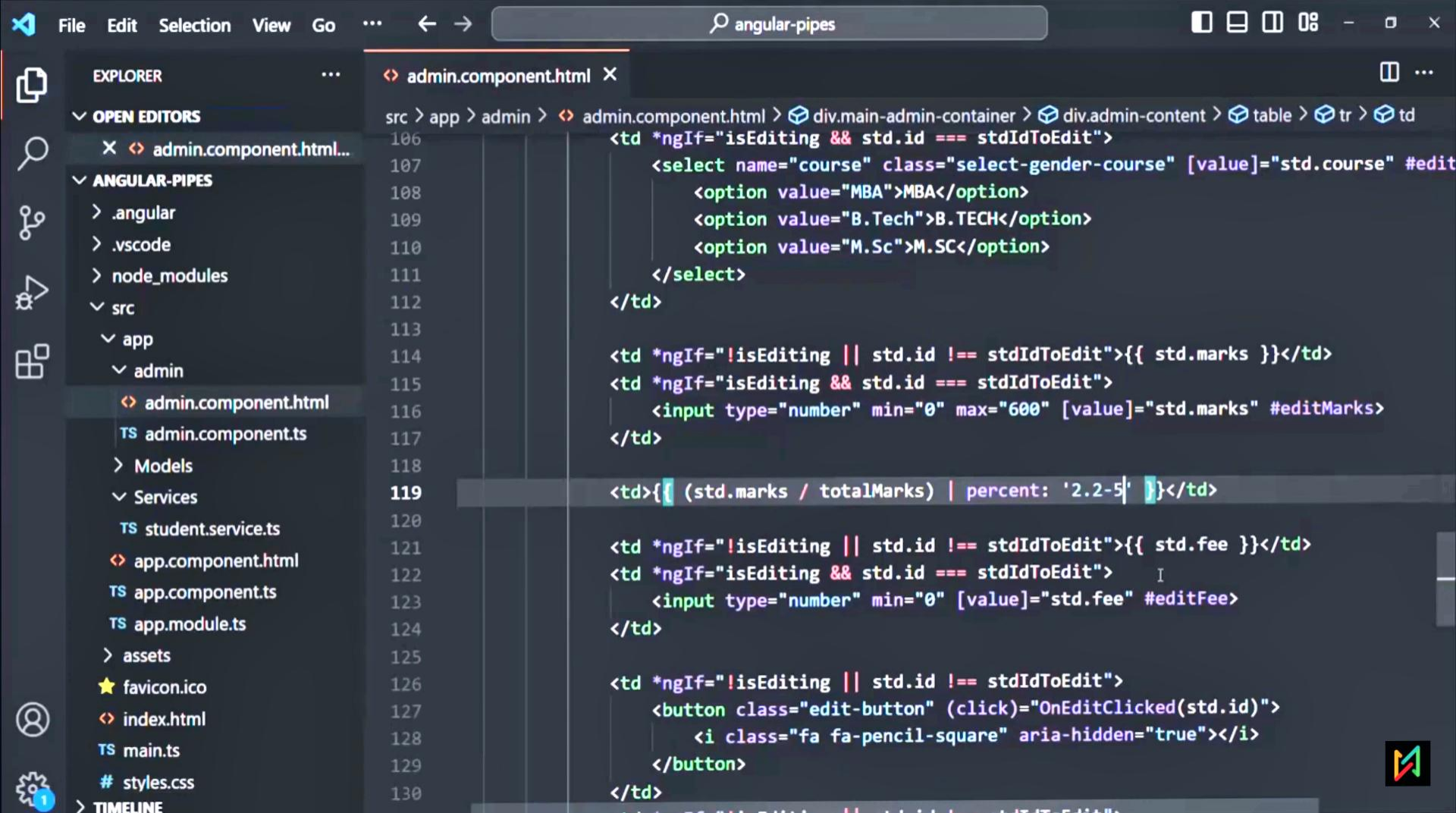
File Edit Selection View Go
angular-pipes
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TIMELINE
src > app > admin > admin.component.html > div.main-admin-container > div.admin-content > table > tr > td
<td *ngIf="isEditing && std.id === stdIdToEdit">
<select name="course" class="select-gender-course" [value]="std.course" #edit
<option value="MBA">MBA</option>
<option value="B.Tech">B.TECH</option>
<option value="M.Sc">M.SC</option>
</select>
</td>
<td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.marks }}</td>
<td *ngIf="isEditing && std.id === stdIdToEdit">
<input type="number" min="0" max="600" [value]="std.marks" #editMarks>
</td>
<td>{{ (std.marks / totalMarks) | percent: '2.2-5' }}</td>
<td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.fee }}</td>
<td *ngIf="isEditing && std.id === stdIdToEdit">
<input type="number" min="0" [value]="std.fee" #editFee>
</td>
<td *ngIf="!isEditing || std.id !== stdIdToEdit">
<button class="edit-button" (click)="OnEditClicked(std.id)">

</button>
</td>

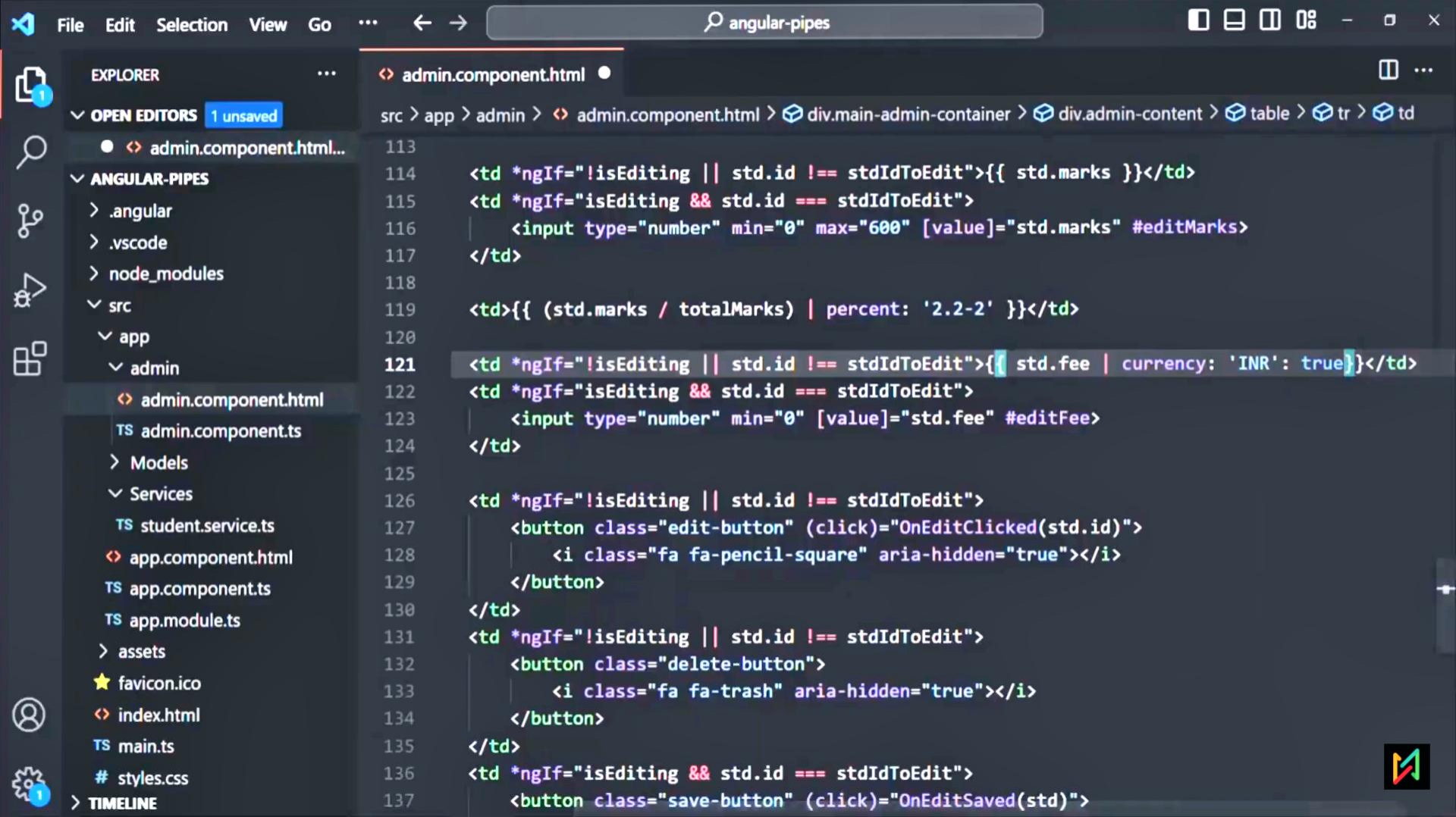
AngularPipes | Angular - API List | Angular - PercentPipe

localhost:4200

Student Management

All

ID	Name	Gender	Date of Birth	Course	Marks	Percentage	Fees		+
1	John Smith	Male	November 12, 1997	MBA	520	86.66667%	1899		
2	Mark Vought	Male	October 6, 1998	B.TECH	420	70.00%	2899		
3	Sarah King	Female	September 22, 1996	B.TECH	540	90.00%	2899		
4	Merry Jane	Female	June 12, 1995	MBA	380	63.33333%	1899		
5	Steve Smith	Male	December 21, 1999	M.SC	430	71.66667%	799		
6	Jonas Weber	Male	June 18, 1997	M.SC	320	53.33333%	799		


File Edit Selection View Go
angular-pipes
EXPLORER
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admin.component.html
src > app > admin > admin.component.html > div.main-admin-container > div.admin-content > table > tr > td
113
114 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.marks }}</td>
115 <td *ngIf="isEditing && std.id === stdIdToEdit">
116 <input type="number" min="0" max="600" [value]="std.marks" #editMarks>
117 </td>
118
119 <td>{{ (std.marks / totalMarks) | percent: '2.2-2' }}</td>
120
121 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.fee | currency: 'INR': true }}</td>
122 <td *ngIf="isEditing && std.id === stdIdToEdit">
123 <input type="number" min="0" [value]="std.fee" #editFee>
124 </td>
125
126 <td *ngIf="!isEditing || std.id !== stdIdToEdit">
127 <button class="edit-button" (click)="OnEditClicked(std.id)">
128
129 </button>
130 </td>
131 <td *ngIf="!isEditing || std.id !== stdIdToEdit">
132 <button class="delete-button">
133
134 </button>
135 </td>
136 <td *ngIf="isEditing && std.id === stdIdToEdit">
137 <button class="save-button" (click)="OnEditSaved(std)">

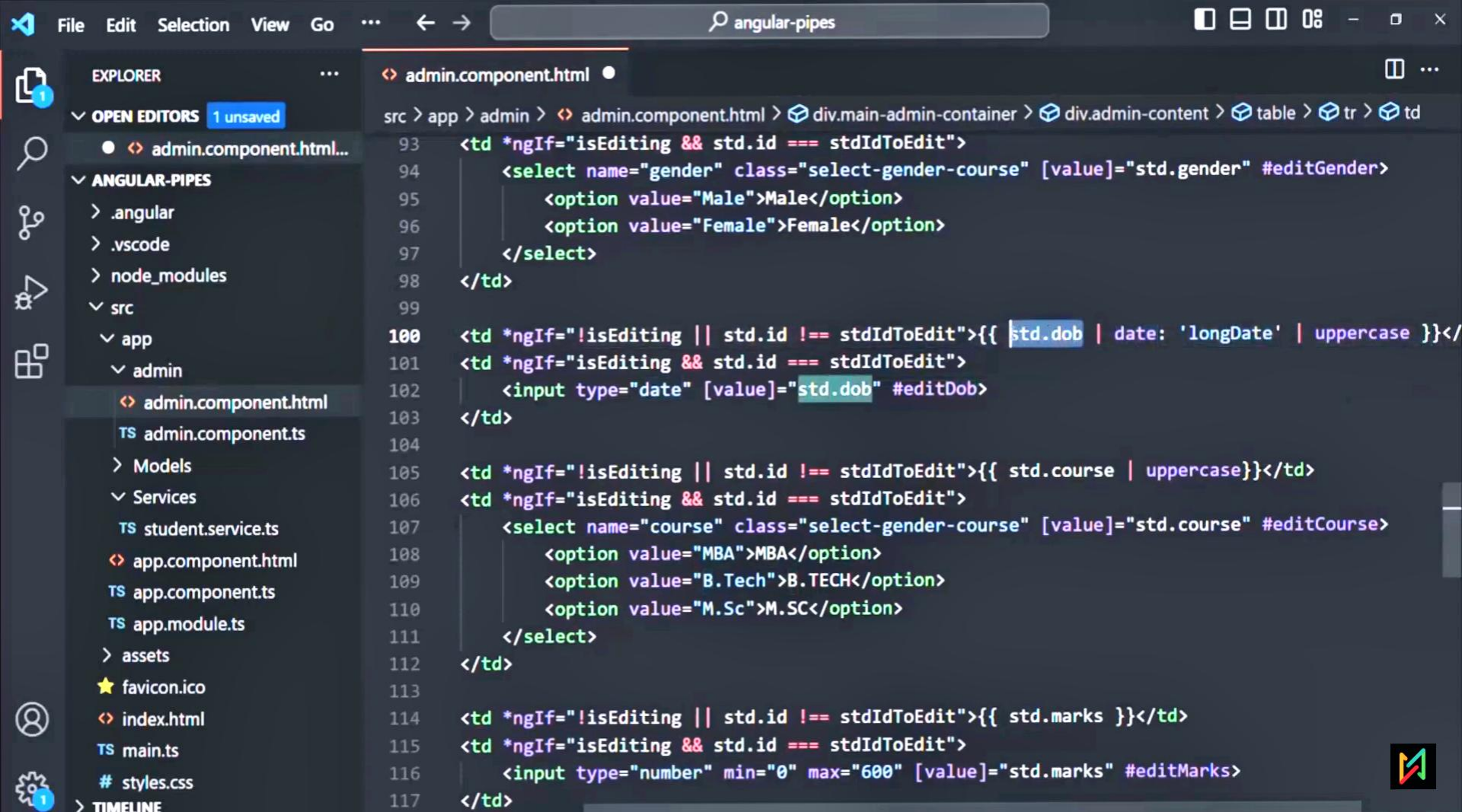
AngularPipes | Angular - API List | Angular - CurrencyPipe

localhost:4200

Student Management

All

ID	Name	Gender	Date of Birth	Course	Marks	Percentage	Fees		+
1	John Smith	Male	November 12, 1997	MBA	520	86.67%	₹1,899.00		
2	Mark Vought	Male	October 6, 1998	B.TECH	420	70.00%	₹2,899.00		
3	Sarah King	Female	September 22, 1996	B.TECH	540	90.00%	₹2,899.00		
4	Merry Jane	Female	June 12, 1995	MBA	380	63.33%	₹1,899.00		
5	Steve Smith	Male	December 21, 1999	M.SC	430	71.67%	₹799.00		
6	Jonas Weber	Male	June 18, 1997	M.SC	320	53.33%	₹799.00		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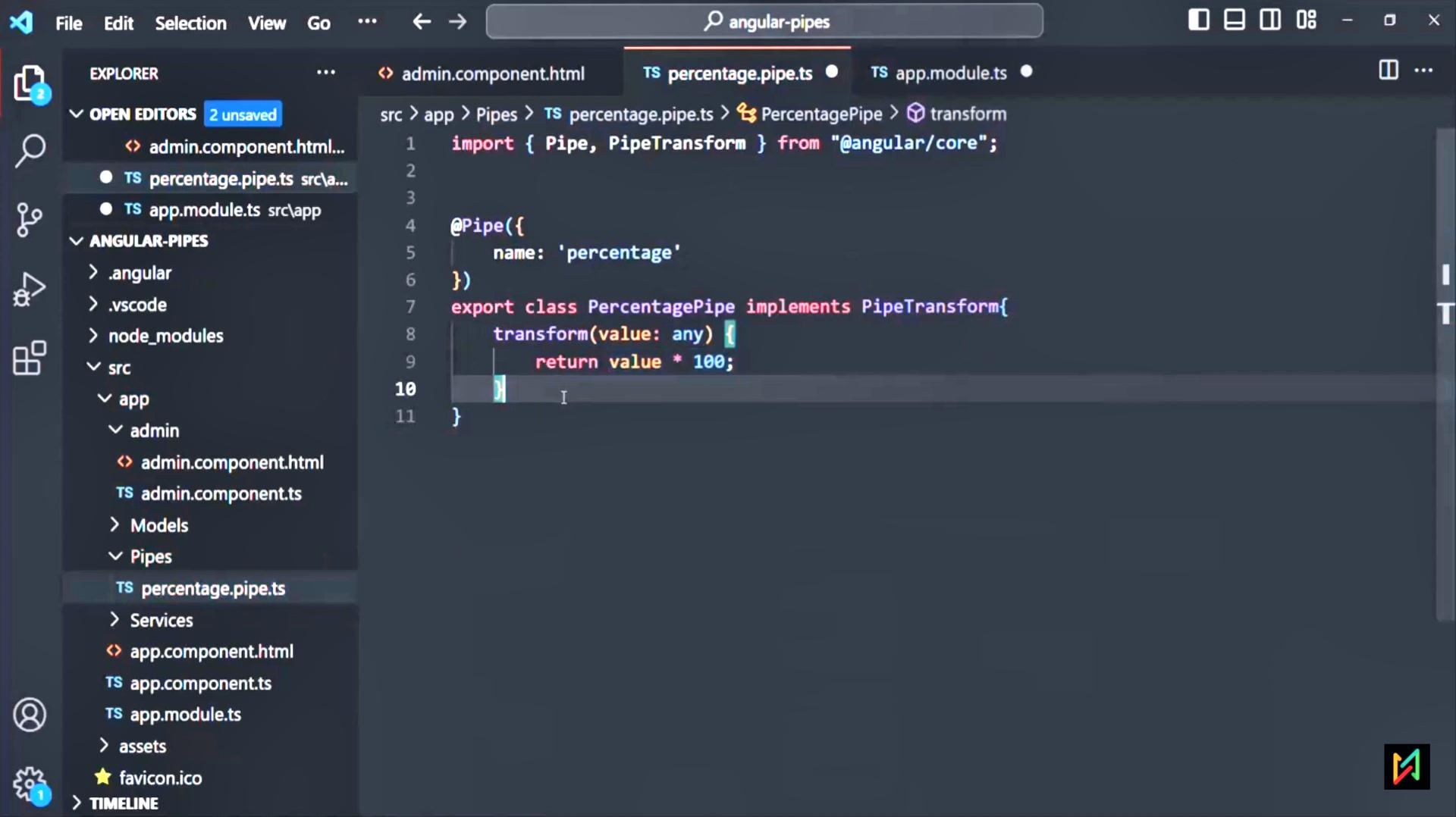
TIMELINE
admin.component.html
src > app > admin > admin.component.html > div.main-admin-container > div.admin-content > table > tr > td
93 <td *ngIf="isEditing && std.id === stdIdToEdit">
94 <select name="gender" class="select-gender-course" [value]="std.gender" #editGender>
95 <option value="Male">Male</option>
96 <option value="Female">Female</option>
97 </select>
98 </td>
99
100 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.dob | date: 'longDate' | uppercase }}</
101 <td *ngIf="isEditing && std.id === stdIdToEdit">
102 <input type="date" [value]="std.dob" #editDob>
103 </td>
104
105 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.course | uppercase}}</td>
106 <td *ngIf="isEditing && std.id === stdIdToEdit">
107 <select name="course" class="select-gender-course" [value]="std.course" #editCourse>
108 <option value="MBA">MBA</option>
109 <option value="B.Tech">B.TECH</option>
110 <option value="M.Sc">M.SC</option>
111 </select>
112 </td>
113
114 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.marks }}</td>
115 <td *ngIf="isEditing && std.id === stdIdToEdit">
116 <input type="number" min="0" max="600" [value]="std.marks" #editMarks>
117 </td>

How to create a custom pipe?

We can create a custom angular pipe in three simple steps:

1. Create a typescript class and export it. By convention, a pipe class name should end with Pipe.
2. Decorate that class with `@Pipe` Decorator. There we can specify a name for the pipe.
3. Inherit `PipeTransform` interface and implement its `transform` method.


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
percentage.pipe.ts
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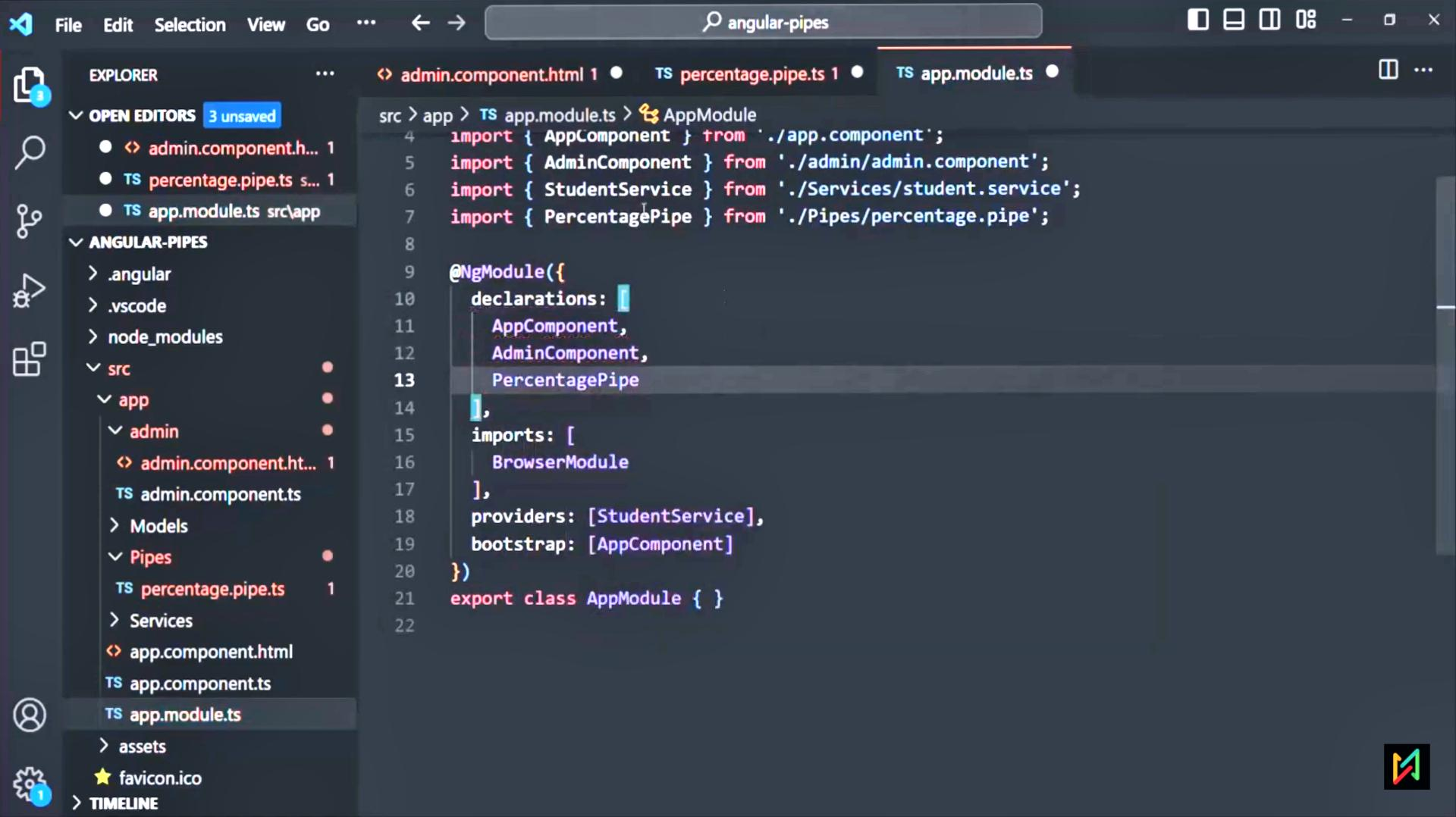
OPEN EDITORS 2 unsaved
admin.component.html...
percentage.pipe.ts src\a...
app.module.ts src\app
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
percentage.pipe.ts
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
TIMELINE
src > app > Pipes > percentage.pipe.ts > PercentagePipe > transform
1 import { Pipe, PipeTransform } from "@angular/core";
2
3
4 @Pipe({
5 name: 'percentage'
6 })
7 export class PercentagePipe implements PipeTransform{
8 transform(value: any) {
9 return value * 100;
10 }
11 }


File Edit Selection View Go
angular-pipes
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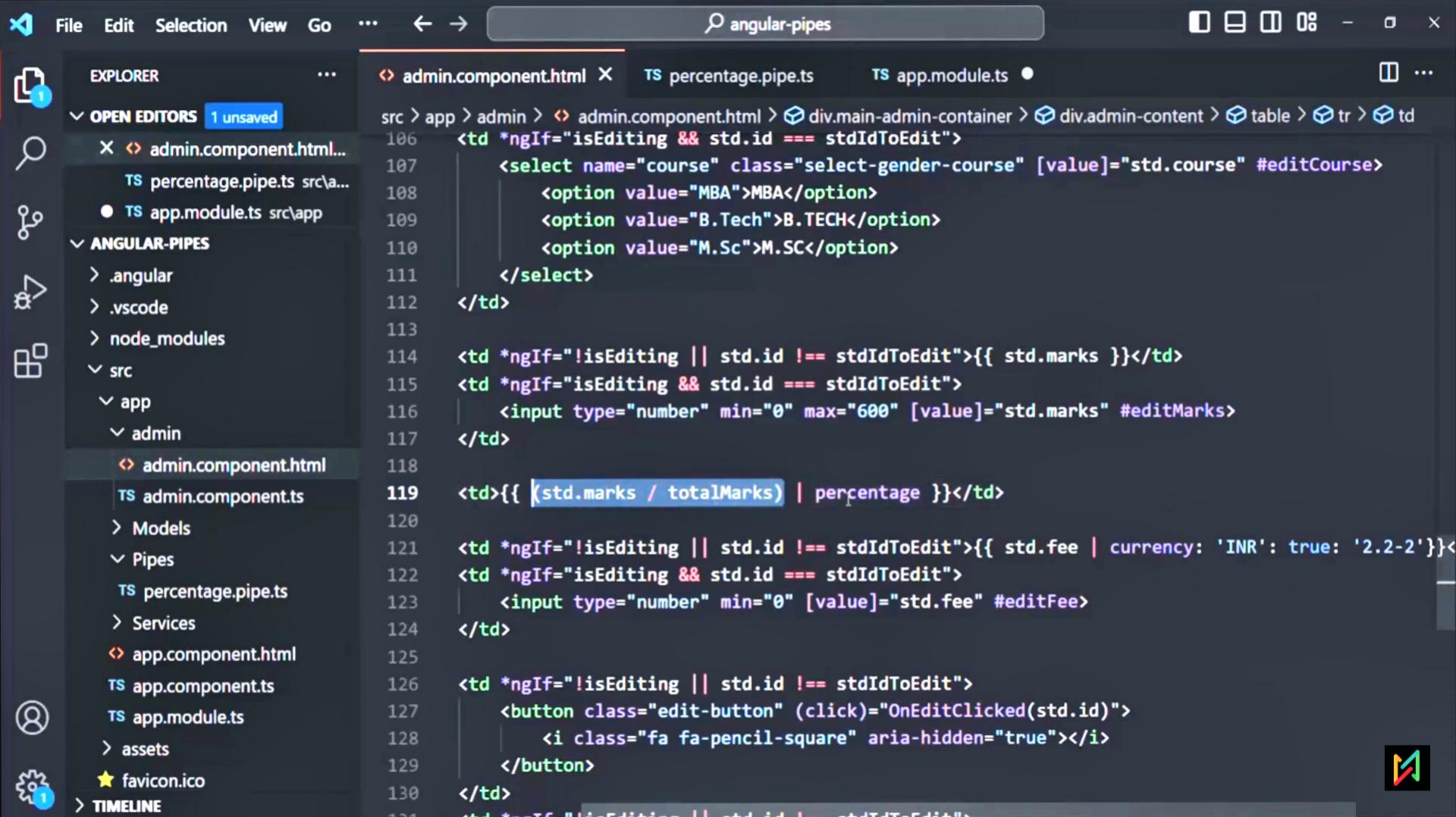
admin.component.ts
Models
Pipes
percentage.pipe.ts 1
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
TIMELINE
src > app > app.module.ts > AppModule
import { AppComponent } from './app.component';
import { AdminComponent } from './admin/admin.component';
import { StudentService } from './Services/student.service';
import { PercentagePipe } from './Pipes/percentage.pipe';
@NgModule({
declarations: [
AppComponent,
AdminComponent,
PercentagePipe
],
imports: [
BrowserModule
],
providers: [StudentService],
bootstrap: [AppComponent]
})
export class AppModule { }


File Edit Selection View Go
angular-pipes
EXPLORER
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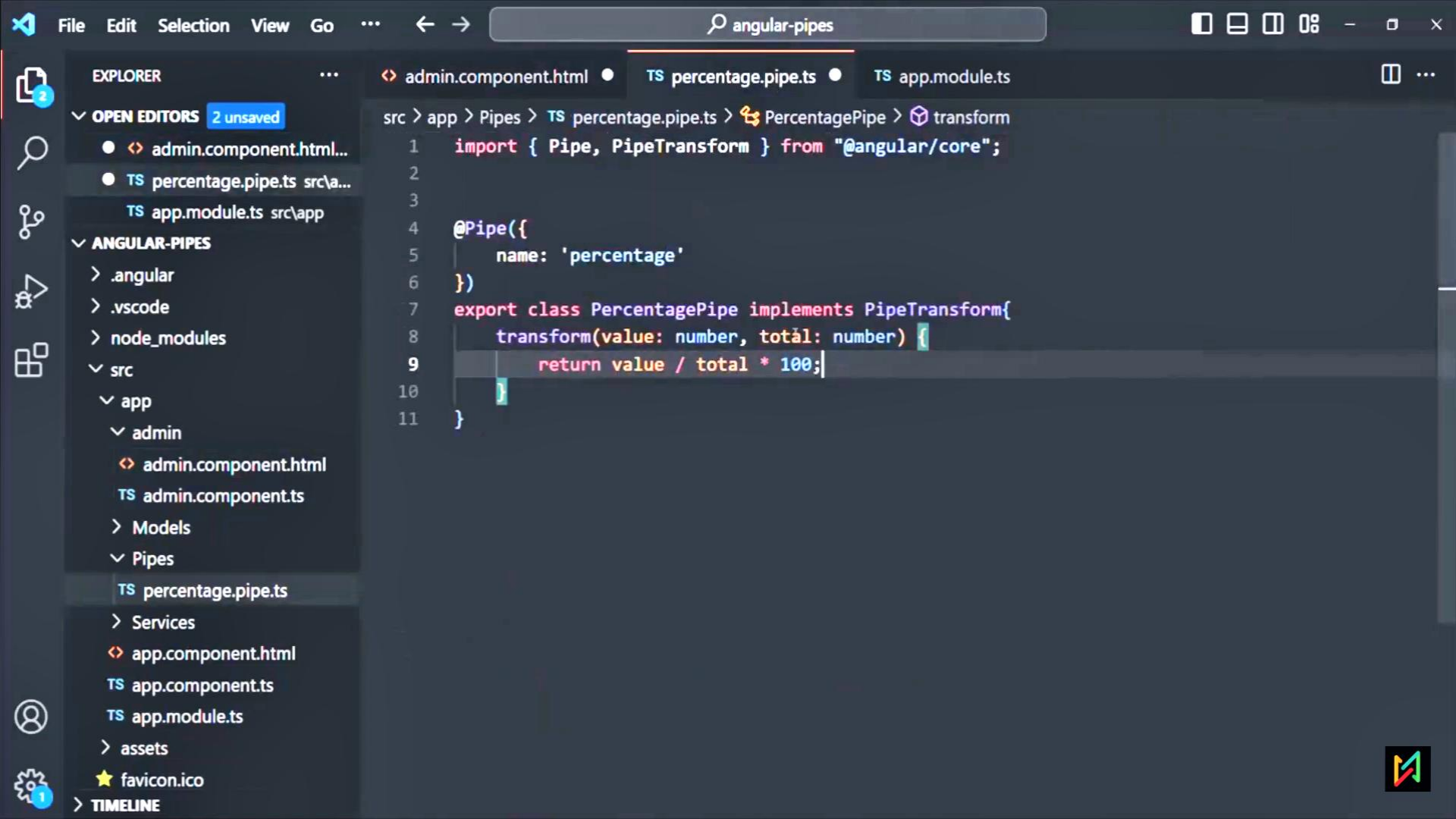
admin.component.ts
Models
Pipes
percentage.pipe.ts
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
TIMELINE
src > app > admin > admin.component.html > div.main-admin-container > div.admin-content > table > tr > td
106 <td *ngIf="isEditing && std.id === stdIdToEdit">
107 <select name="course" class="select-gender-course" [value]="std.course" #editCourse>
108 <option value="MBA">MBA</option>
109 <option value="B.Tech">B.TECH</option>
110 <option value="M.Sc">M.SC</option>
111 </select>
112 </td>
113
114 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.marks }}</td>
115 <td *ngIf="isEditing && std.id === stdIdToEdit">
116 <input type="number" min="0" max="600" [value]="std.marks" #editMarks>
117 </td>
118
119 <td>{{ (std.marks / totalMarks) | percentage }}</td>
120
121 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.fee | currency: 'INR': true: '2.2-2'}}
122 <td *ngIf="isEditing && std.id === stdIdToEdit">
123 <input type="number" min="0" [value]="std.fee" #editFee>
124 </td>
125
126 <td *ngIf="!isEditing || std.id !== stdIdToEdit">
127 <button class="edit-button" (click)="OnEditClicked(std.id)">
128
129 </button>
130 </td>


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
percentage.pipe.ts
app.module.ts
src > app > Pipes > percentage.pipe.ts > PercentagePipe > transform
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admin.component.html...
percentage.pipe.ts src\a...
app.module.ts src\app
ANGULAR-PIPES
.angular
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node_modules
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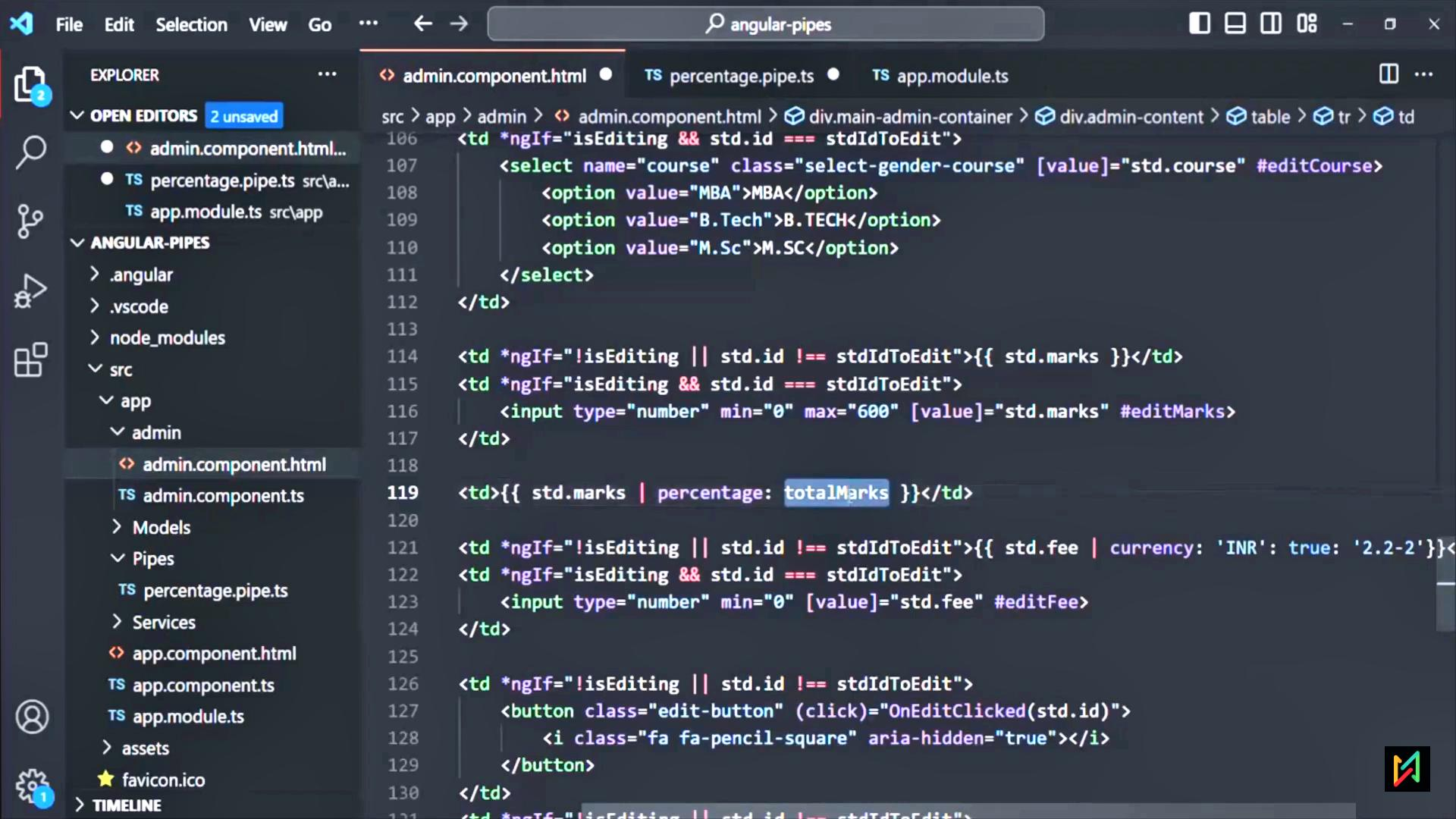
admin.component.html
admin.component.ts
Models
Pipes
percentage.pipe.ts
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
TIMELINE
import { Pipe, PipeTransform } from "@angular/core";
@Pipe({
name: 'percentage'
})
export class PercentagePipe implements PipeTransform{
transform(value: number, total: number) {
return value / total * 100;
}
}


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
percentage.pipe.ts
app.module.ts
src > app > admin > admin.component.html > div.main-admin-container > div.admin-content > table > tr > td
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percentage.pipe.ts src\a...
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ANGULAR-PIPES
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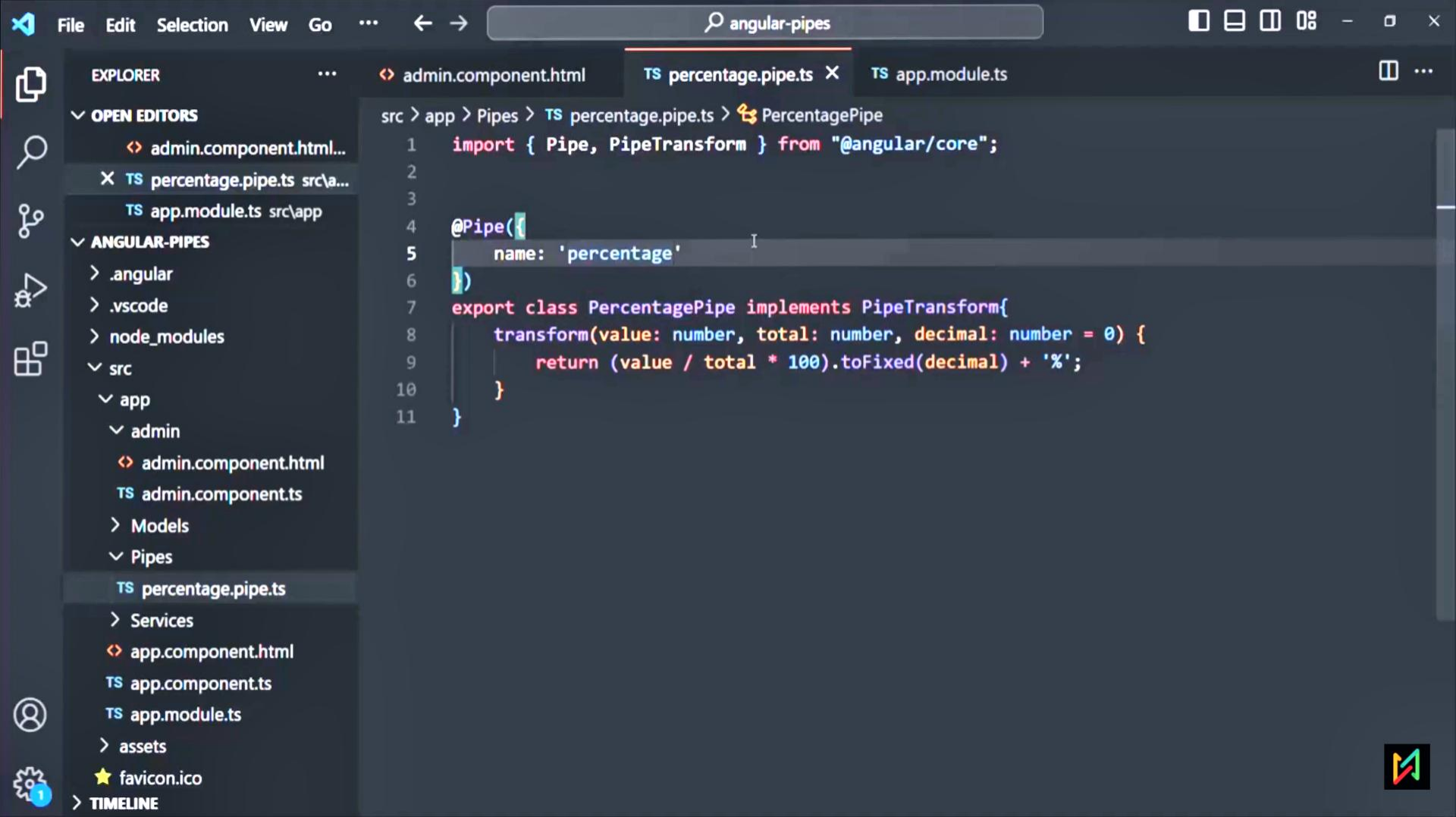
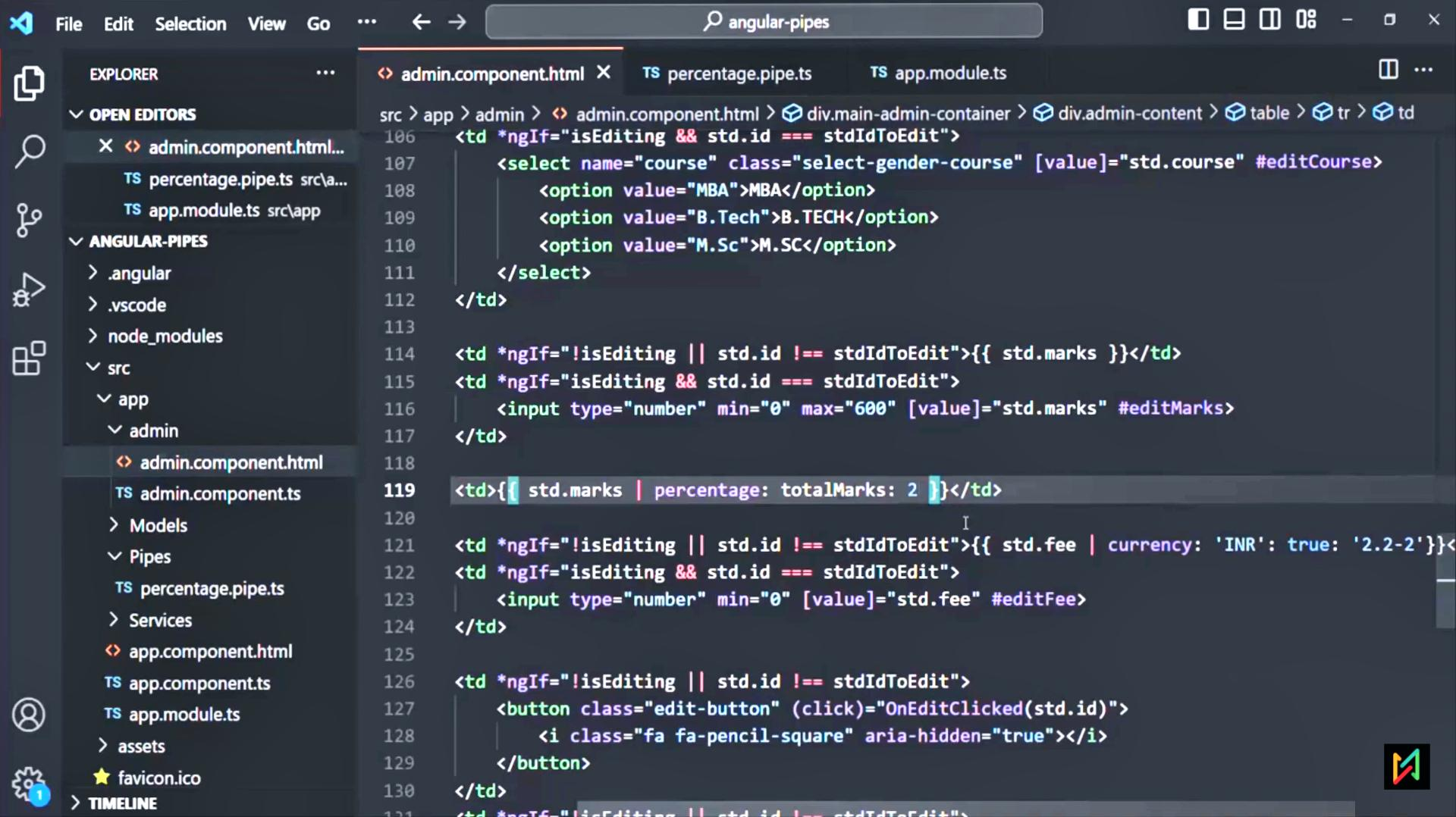
Models
Pipes
percentage.pipe.ts
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
TIMELINE
106 <td *ngIf="isEditing && std.id === stdIdToEdit">
107 <select name="course" class="select-gender-course" [value]="std.course" #editCourse>
108 <option value="MBA">MBA</option>
109 <option value="B.Tech">B.TECH</option>
110 <option value="M.Sc">M.SC</option>
111 </select>
112 </td>
113
114 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.marks }}</td>
115 <td *ngIf="isEditing && std.id === stdIdToEdit">
116 <input type="number" min="0" max="600" [value]="std.marks" #editMarks>
117 </td>
118
119 <td>{{ std.marks | percentage: totalMarks }}</td>
120
121 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.fee | currency: 'INR': true: '2.2-2'}}
122 <td *ngIf="isEditing && std.id === stdIdToEdit">
123 <input type="number" min="0" [value]="std.fee" #editFee>
124 </td>
125
126 <td *ngIf="!isEditing || std.id !== stdIdToEdit">
127 <button class="edit-button" (click)="OnEditClicked(std.id)">
128
129 </button>
130 </td>


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
percentage.pipe.ts
app.module.ts
src > app > Pipes > percentage.pipe.ts > PercentagePipe
OPEN EDITORS
admin.component.html...
percentage.pipe.ts src\a...
app.module.ts src\app
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
percentage.pipe.ts
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
TIMELINE
1 import { Pipe, PipeTransform } from "@angular/core";
2
3
4 @Pipe({
5 name: 'percentage'
6 })
7 export class PercentagePipe implements PipeTransform{
8 transform(value: number, total: number, decimal: number = 0) {
9 return (value / total * 100).toFixed(decimal) + '%';
10 }
11 }


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
percentage.pipe.ts
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OPEN EDITORS
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percentage.pipe.ts src\a...
app.module.ts src\app
ANGULAR-PIPES
.angular
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node_modules
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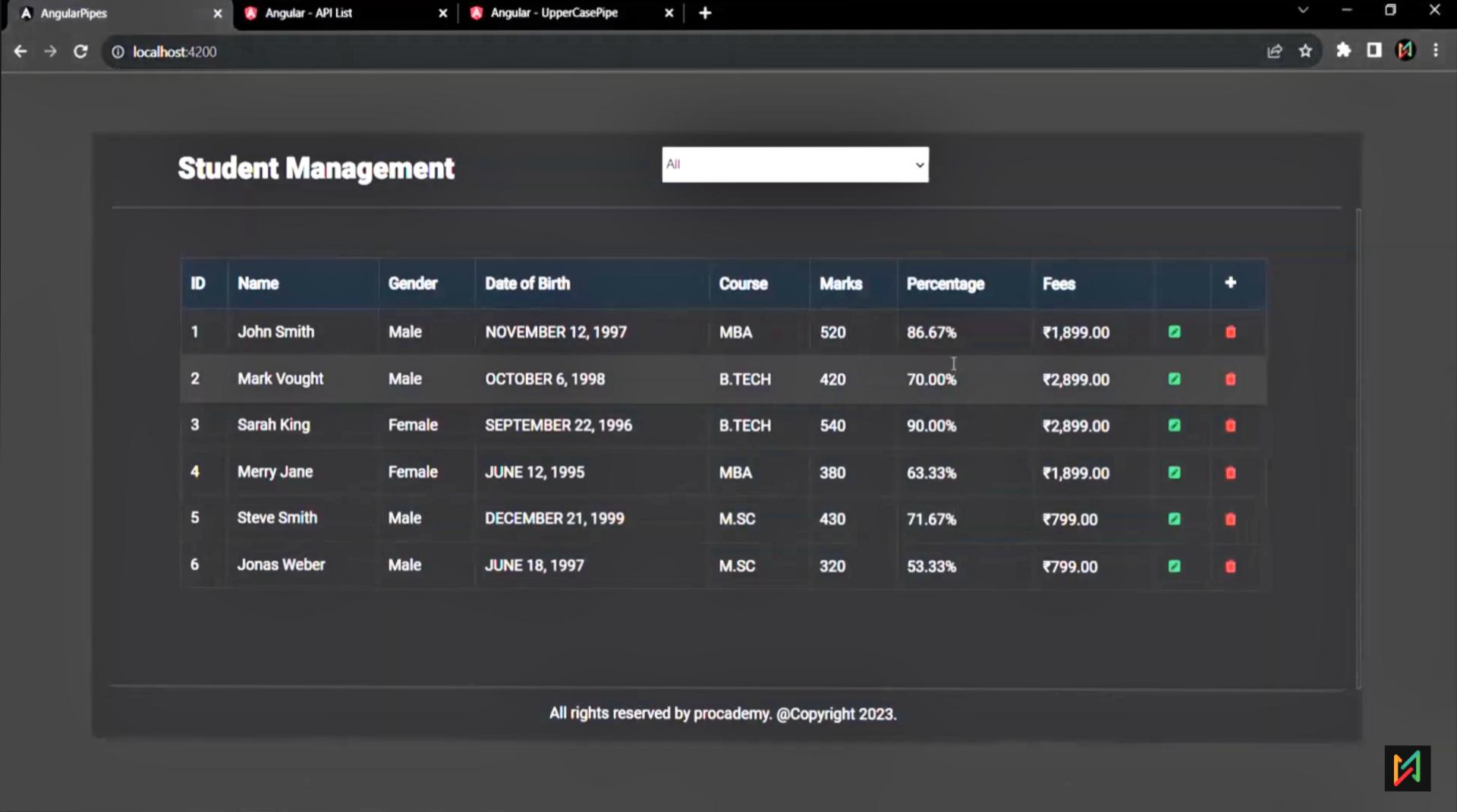
admin
admin.component.html
admin.component.ts
Models
Pipes
percentage.pipe.ts
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
TIMELINE
src > app > admin > admin.component.html > div.main-admin-container > div.admin-content > table > tr > td
106 <td *ngIf="isEditing && std.id === stdIdToEdit">
107 <select name="course" class="select-gender-course" [value]="std.course" #editCourse>
108 <option value="MBA">MBA</option>
109 <option value="B.Tech">B.TECH</option>
110 <option value="M.Sc">M.SC</option>
111 </select>
112 </td>
113
114 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.marks }}</td>
115 <td *ngIf="isEditing && std.id === stdIdToEdit">
116 <input type="number" min="0" max="600" [value]="std.marks" #editMarks>
117 </td>
118
119 <td>{{ std.marks | percentage: totalMarks: 2 }}</td>
120
121 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.fee | currency: 'INR': true: '2.2-2'}}
122 <td *ngIf="isEditing && std.id === stdIdToEdit">
123 <input type="number" min="0" [value]="std.fee" #editFee>
124 </td>
125
126 <td *ngIf="!isEditing || std.id !== stdIdToEdit">
127 <button class="edit-button" (click)="OnEditClicked(std.id)">
128
129 </button>
130 </td>


AngularPipes
Angular - API List
Angular - UpperCasePipe
localhost:4200
Student Management
All
ID
Name
Gender
Date of Birth
Course
Marks
Percentage
Fees
1
John Smith
Male
NOVEMBER 12, 1997
MBA
520
86.67%
₹1,899.00
2
Mark Vought
Male
OCTOBER 6, 1998
B.TECH
420
70.00%
₹2,899.00
3
Sarah King
Female
SEPTEMBER 22, 1996
B.TECH
540
90.00%
₹2,899.00
4
Merry Jane
Female
JUNE 12, 1995
MBA
380
63.33%
₹1,899.00
5
Steve Smith
Male
DECEMBER 21, 1999
M.SC
430
71.67%
₹799.00
6
Jonas Weber
Male
JUNE 18, 1997
M.SC
320
53.33%
₹799.00
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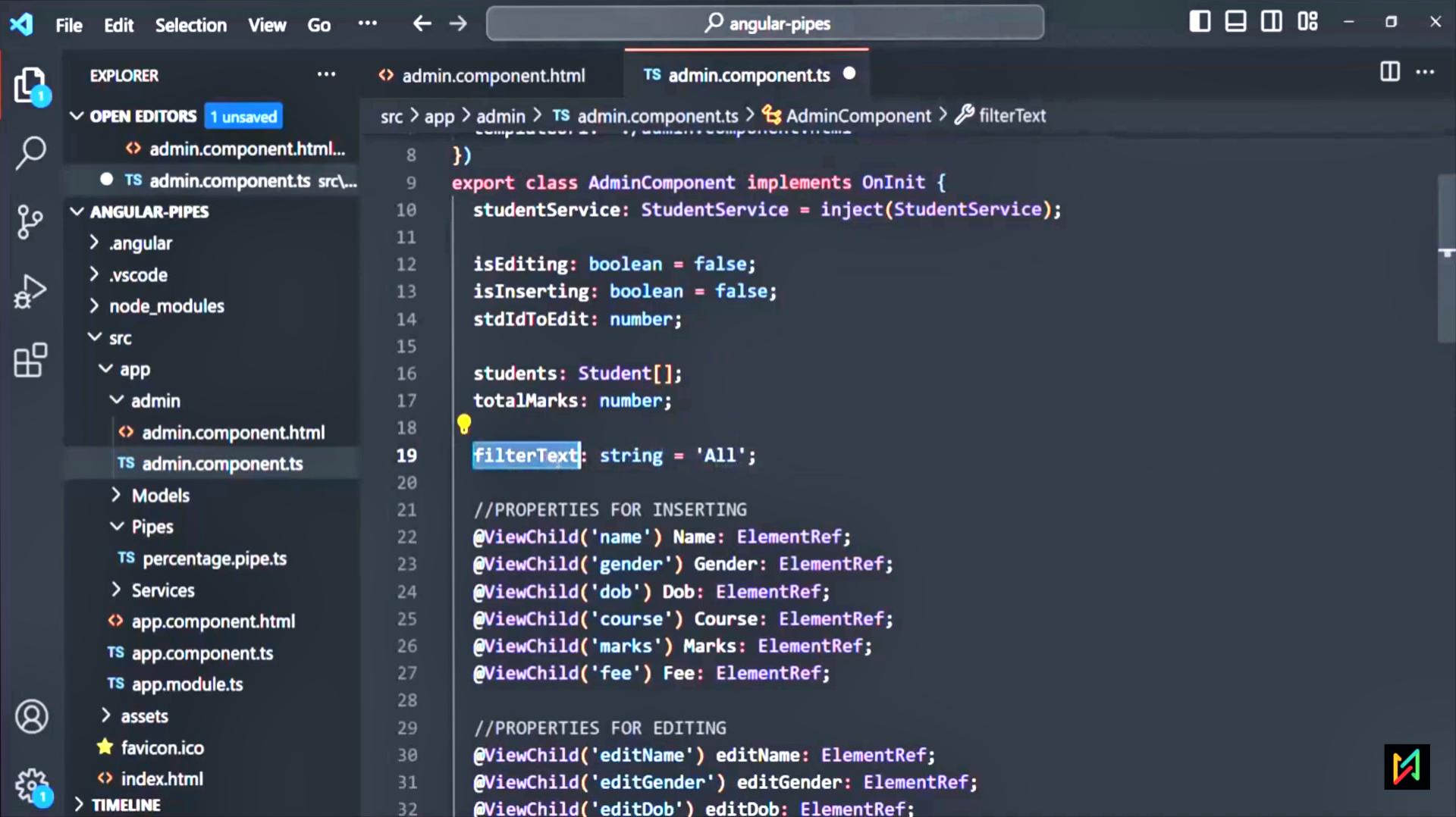
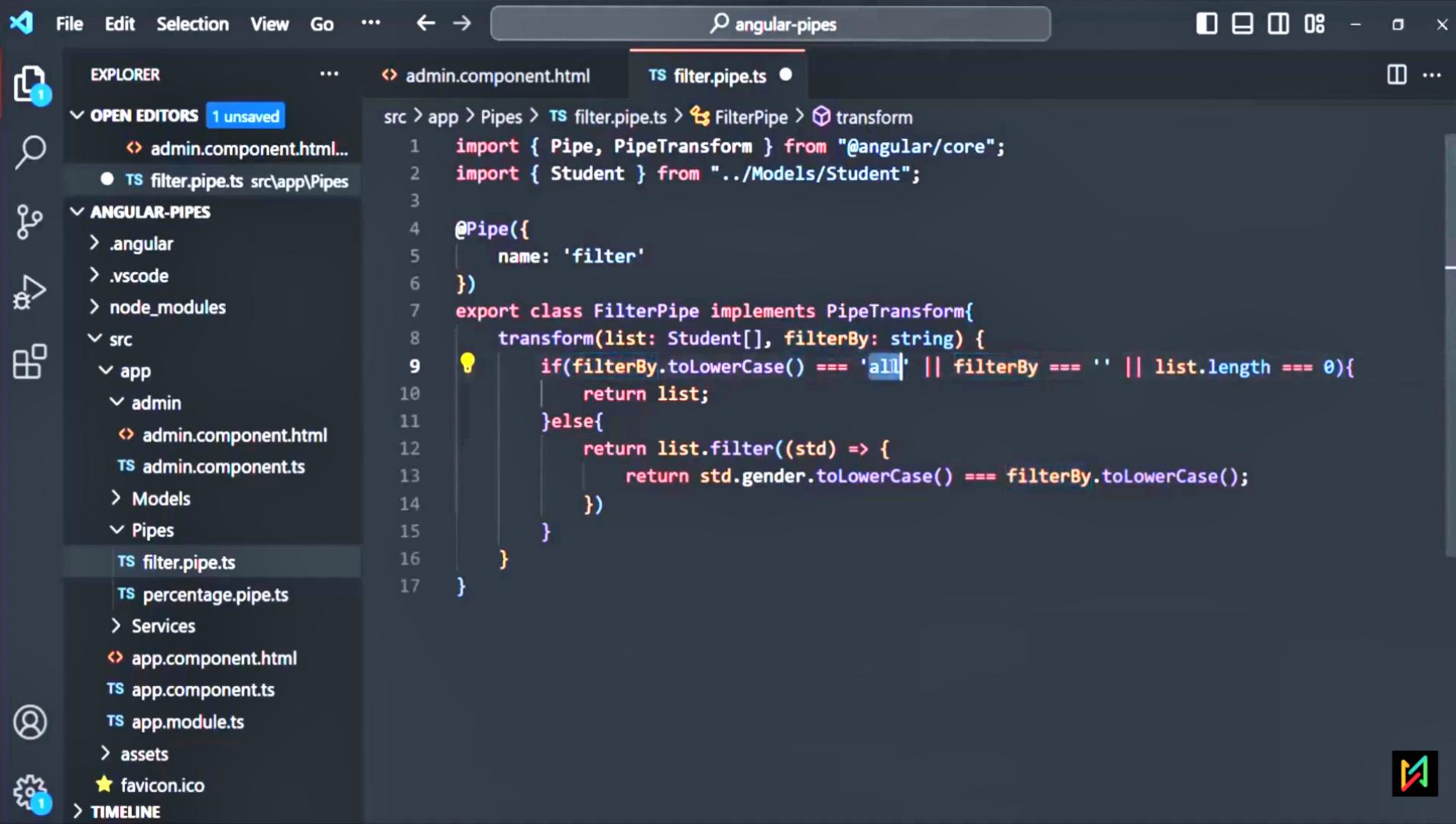
Creating a Custom Filter Pipes


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
admin.component.ts
OPEN EDITORS 1 unsaved
admin.component.html...
admin.component.ts src\...
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
percentage.pipe.ts
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
index.html
TIMELINE
src > app > admin > admin.component.ts > AdminComponent > filterText
8 })
9 export class AdminComponent implements OnInit {
10 studentService: StudentService = inject(StudentService);
11
12 isEditing: boolean = false;
13 isInserting: boolean = false;
14 stdIdToEdit: number;
15
16 students: Student[];
17 totalMarks: number;
18
19 filterText: string = 'All';
20
21 //PROPERTIES FOR INSERTING
22 @ViewChild('name') Name: ElementRef;
23 @ViewChild('gender') Gender: ElementRef;
24 @ViewChild('dob') Dob: ElementRef;
25 @ViewChild('course') Course: ElementRef;
26 @ViewChild('marks') Marks: ElementRef;
27 @ViewChild('fee') Fee: ElementRef;
28
29 //PROPERTIES FOR EDITING
30 @ViewChild('editName') editName: ElementRef;
31 @ViewChild('editGender') editGender: ElementRef;
32 @ViewChild('editDob') editDob: ElementRef;


File Edit Selection View Go
angular-pipes
EXPLORER
OPEN EDITORS 1 unsaved
admin.component.html...
filter.pipe.ts src\app\Pipes
ANGULAR-PIPES
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src
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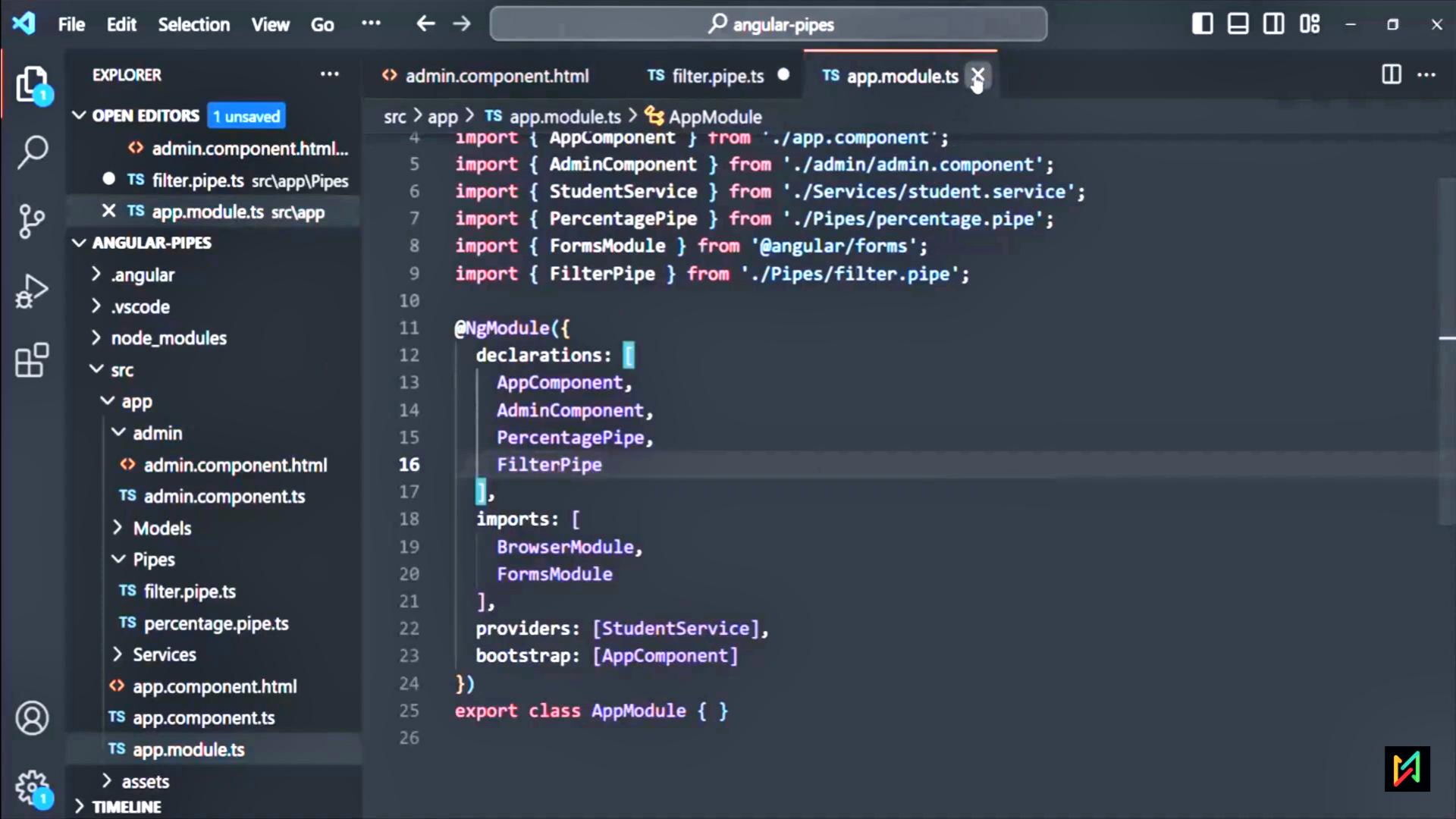
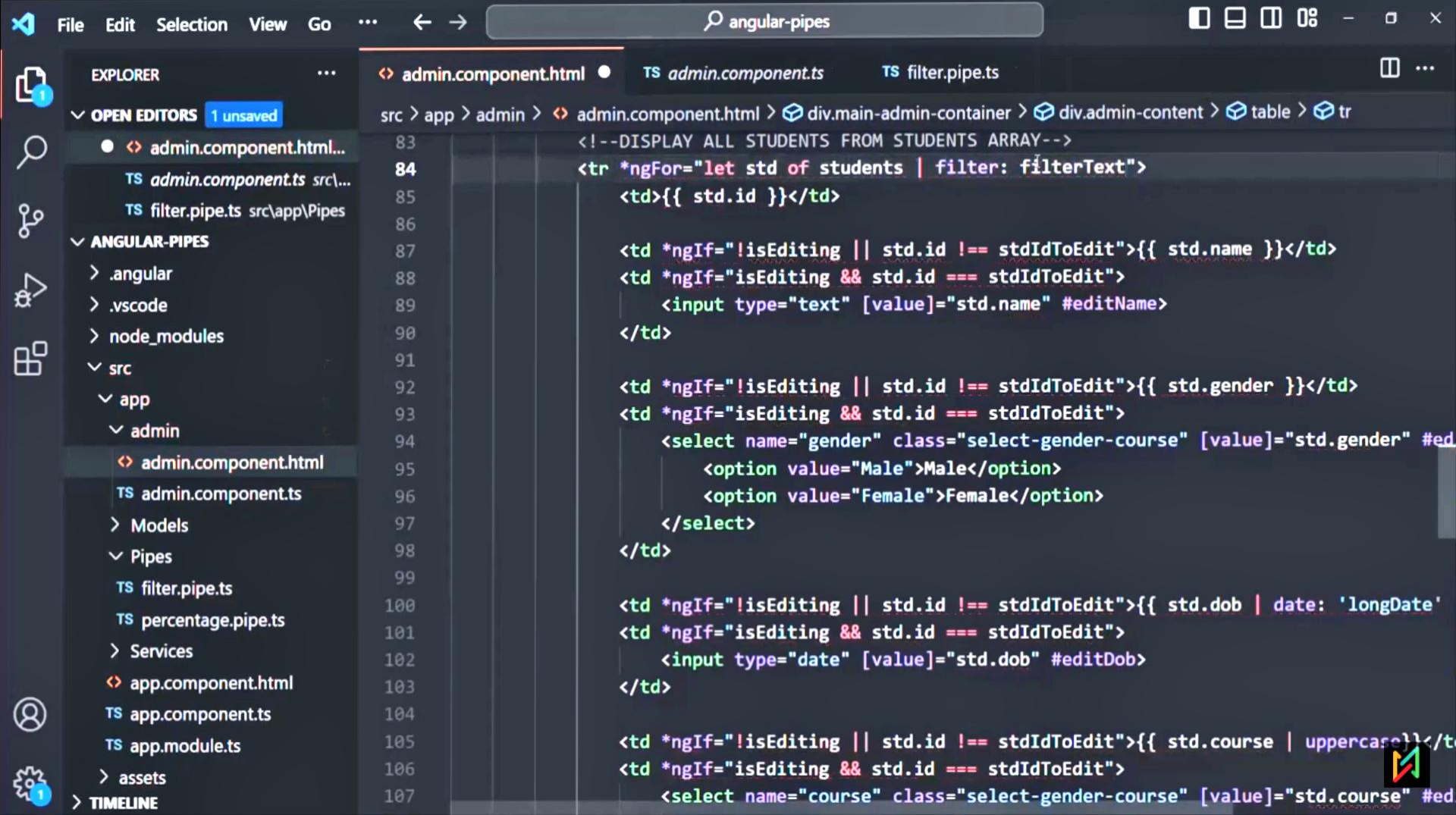
admin.component.html
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
TIMELINE
admin.component.html
filter.pipe.ts
src > app > Pipes > filter.pipe.ts > FilterPipe > transform
import { Pipe, PipeTransform } from "@angular/core";
import { Student } from "../Models/Student";
@Pipe({
name: 'filter'
})
export class FilterPipe implements PipeTransform{
transform(list: Student[], filterBy: string) {
if(filterBy.toLowerCase() === 'all' || filterBy === '' || list.length === 0){
return list;
}else{
return list.filter((std) => {
return std.gender.toLowerCase() === filterBy.toLowerCase();
})
}
}
}


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
filter.pipe.ts
app.module.ts
src > app > app.module.ts > AppModule
OPEN EDITORS 1 unsaved
admin.component.html...
filter.pipe.ts src\app\Pipes
app.module.ts src\app
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
app.component.html
app.component.ts
app.module.ts
assets
TIMELINE
import { AppComponent } from './app.component';
import { AdminComponent } from './admin/admin.component';
import { StudentService } from './Services/student.service';
import { PercentagePipe } from './Pipes/percentage.pipe';
import { FormsModule } from '@angular/forms';
import { FilterPipe } from './Pipes/filter.pipe';
@NgModule({
declarations: [
AppComponent,
AdminComponent,
PercentagePipe,
FilterPipe
],
imports: [
BrowserModule,
FormsModule
],
providers: [StudentService],
bootstrap: [AppComponent]
})
export class AppModule { }


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
admin.component.ts
filter.pipe.ts
src > app > admin > admin.component.html > div.main-admin-container > div.admin-content > table > tr
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admin.component.html...
admin.component.ts src\...
filter.pipe.ts src\app\Pipes
ANGULAR-PIPES
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.vscode
node_modules
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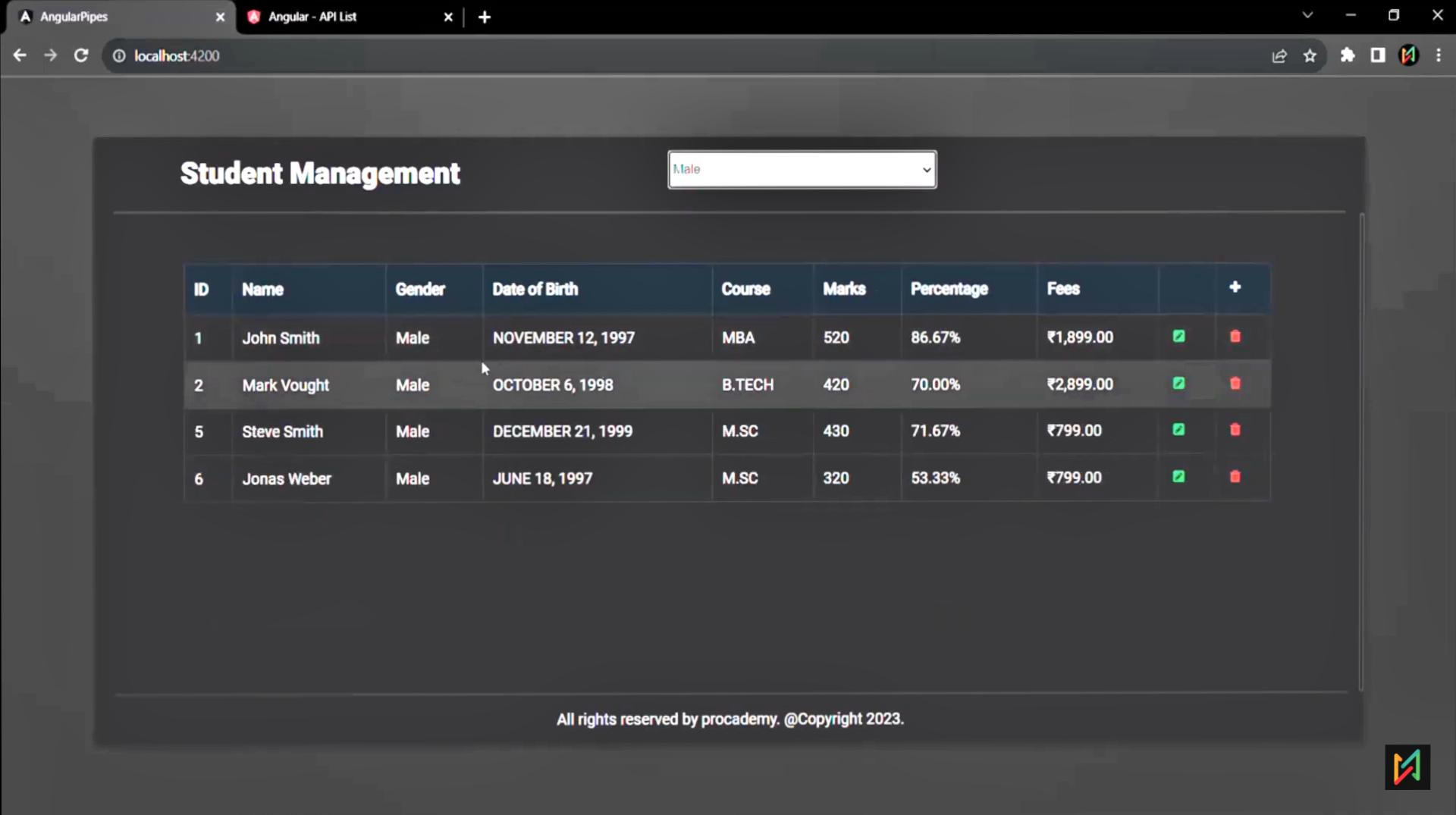
admin.component.html
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
app.component.html
app.component.ts
app.module.ts
assets
TIMELINE
83 <!--DISPLAY ALL STUDENTS FROM STUDENTS ARRAY-->
84 <tr *ngFor="let std of students | filter: filterText">
85 <td>{{ std.id }}</td>
86
87 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.name }}</td>
88 <td *ngIf="isEditing && std.id === stdIdToEdit">
89 <input type="text" [value]="std.name" #editName>
90 </td>
91
92 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.gender }}</td>
93 <td *ngIf="isEditing && std.id === stdIdToEdit">
94 <select name="gender" class="select-gender-course" [value]="std.gender" #ed
95 <option value="Male">Male</option>
96 <option value="Female">Female</option>
97 </select>
98 </td>
99
100 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.dob | date: 'longDate'
101 <td *ngIf="isEditing && std.id === stdIdToEdit">
102 <input type="date" [value]="std.dob" #editDob>
103 </td>
104
105 <td *ngIf="!isEditing || std.id !== stdIdToEdit">{{ std.course | uppercase}}</t
106 <td *ngIf="isEditing && std.id === stdIdToEdit">
107 <select name="course" class="select-gender-course" [value]="std.course" #ed


AngularPipes
Angular - API List
localhost:4200
Student Management
Male
ID
Name
Gender
Date of Birth
Course
Marks
Percentage
Fees
1
John Smith
Male
NOVEMBER 12, 1997
MBA
520
86.67%
₹1,899.00
2
Mark Vought
Male
OCTOBER 6, 1998
B.TECH
420
70.00%
₹2,899.00
5
Steve Smith
Male
DECEMBER 21, 1999
M.SC
430
71.67%
₹799.00
6
Jonas Weber
Male
JUNE 18, 1997
M.SC
320
53.33%
₹799.00
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Why to not use pipes for filtering or sorting?

Angular team always recommends not to use pipe to filter or sort data. When we use pipe for filtering or sorting data, it can significantly impact the performance of the application, if not implemented carefully.

Pure & Impure Pipes in Angular

What is a pure pipe?

A pure pipe is that pipe which gets called whenever there is a pure change on the input value. By default every pipe is a pure pipe.

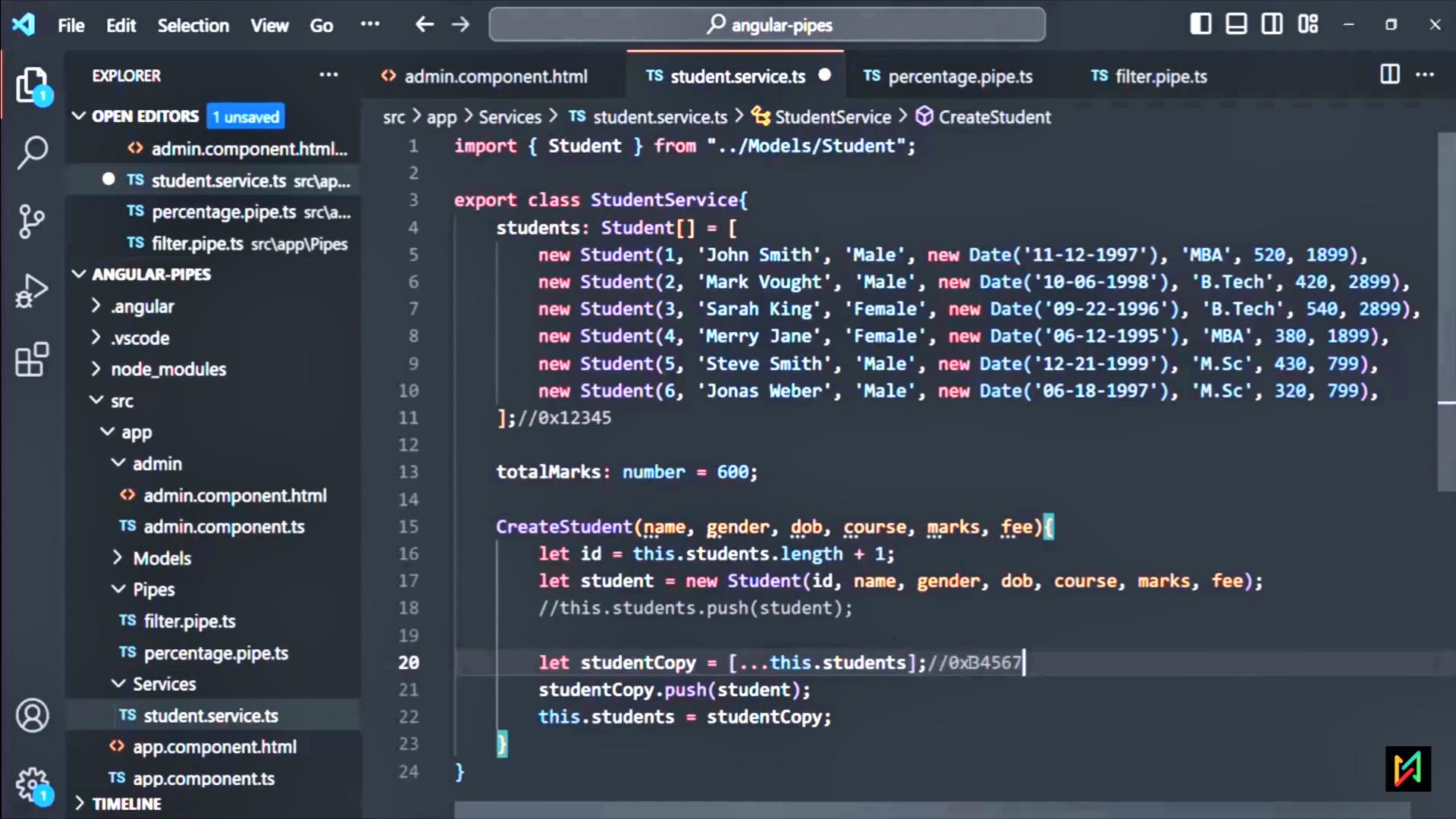
What is pure change?

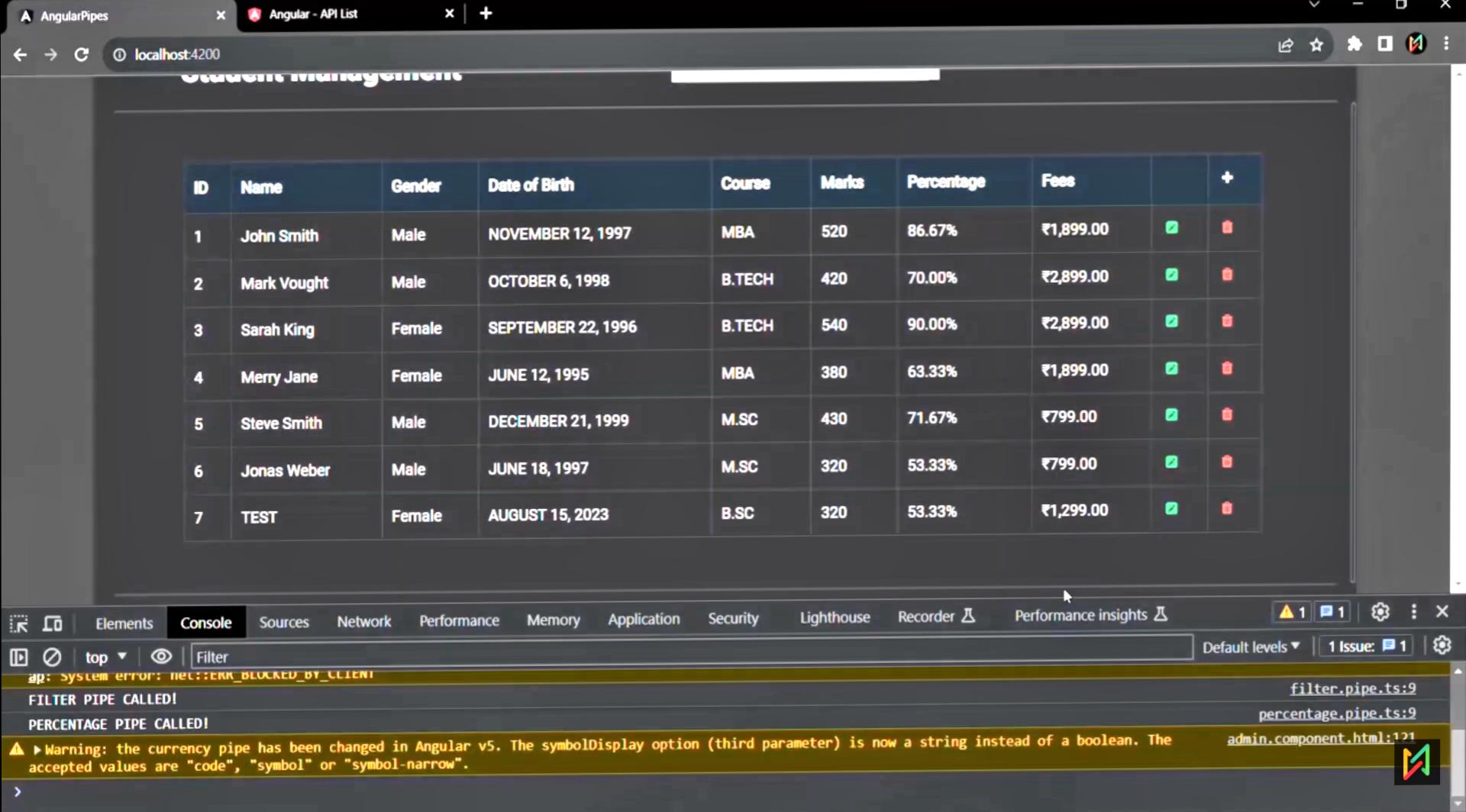
Following changes are considered as pure change in a data:

1. If the pipe is being used on a primitive type like string, number, boolean etc. And if the value of that primitive type changes, it is considered as a pure change

2. If the pipe is being used on a reference type input, and if the reference of that input changes, than the change is pure change.

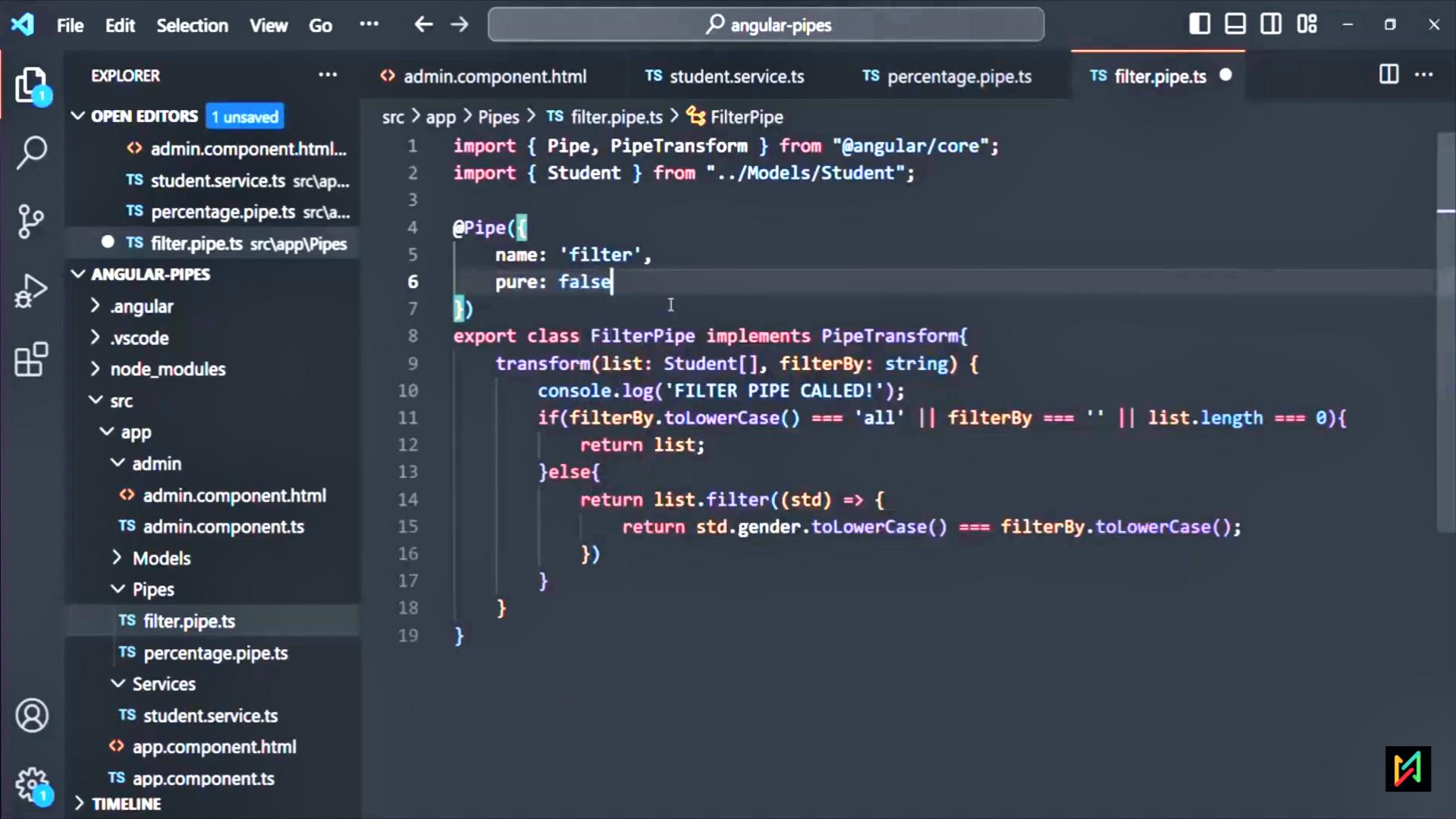
3. **NOTE**: But if the input is of reference type, and only its property value changes and its reference has not changed, that change is not a pure change.


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
student.service.ts
percentage.pipe.ts
filter.pipe.ts
OPEN EDITORS 1 unsaved
admin.component.html...
student.service.ts src\ap...
percentage.pipe.ts src\a...
filter.pipe.ts src\app\Pipes
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
student.service.ts
app.component.html
app.component.ts
TIMELINE
src > app > Services > student.service.ts > StudentService > CreateStudent
import { Student } from "../Models/Student";
export class StudentService{
students: Student[] = [
new Student(1, 'John Smith', 'Male', new Date('11-12-1997'), 'MBA', 520, 1899),
new Student(2, 'Mark Vought', 'Male', new Date('10-06-1998'), 'B.Tech', 420, 2899),
new Student(3, 'Sarah King', 'Female', new Date('09-22-1996'), 'B.Tech', 540, 2899),
new Student(4, 'Merry Jane', 'Female', new Date('06-12-1995'), 'MBA', 380, 1899),
new Student(5, 'Steve Smith', 'Male', new Date('12-21-1999'), 'M.Sc', 430, 799),
new Student(6, 'Jonas Weber', 'Male', new Date('06-18-1997'), 'M.Sc', 320, 799),
];//0x12345
totalMarks: number = 600;
CreateStudent(name, gender, dob, course, marks, fee){
let id = this.students.length + 1;
let student = new Student(id, name, gender, dob, course, marks, fee);
//this.students.push(student);
let studentCopy = [...this.students];//0xB4567
studentCopy.push(student);
this.students = studentCopy;
}
}

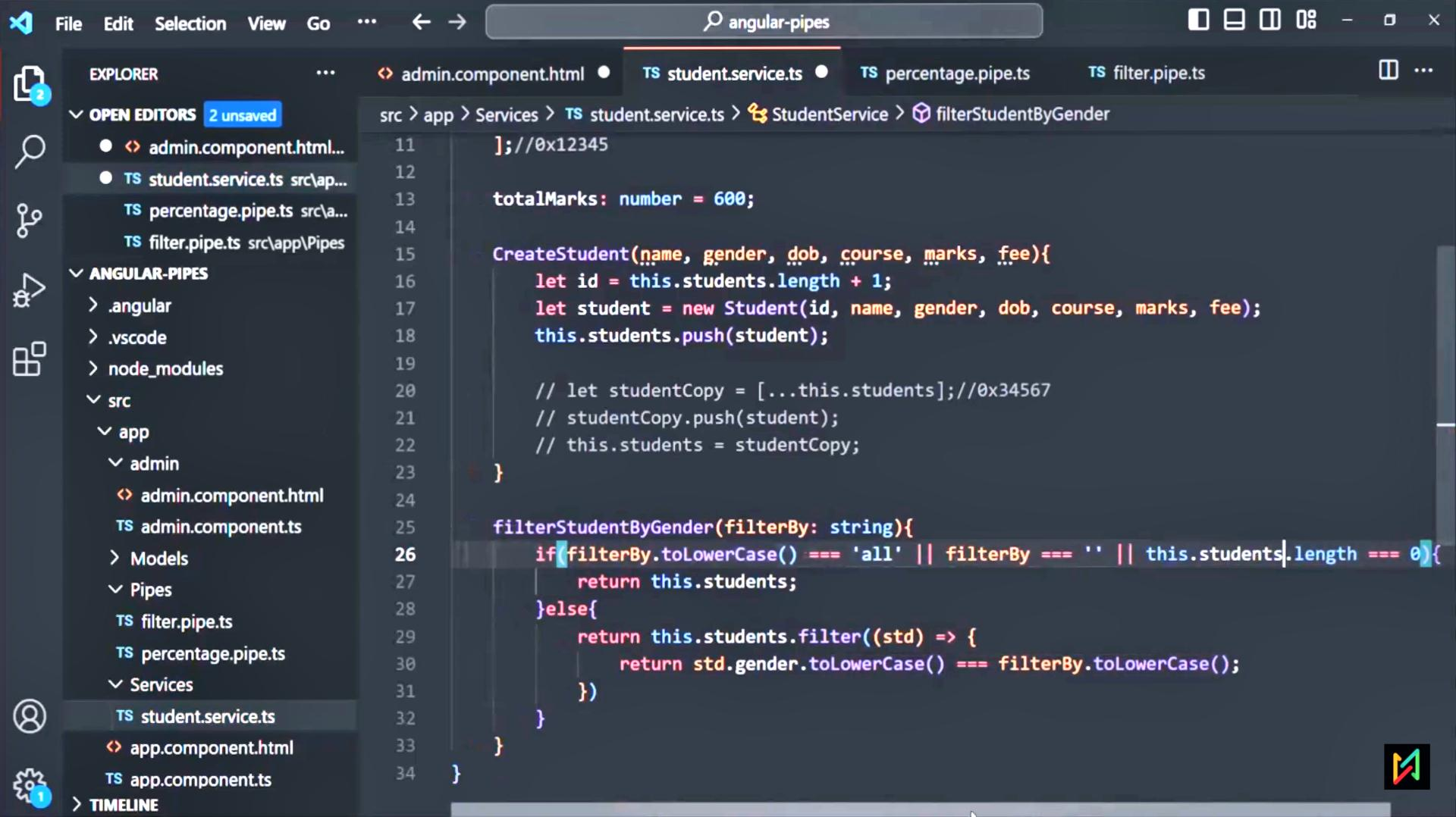

AngularPipes
Angular - API List
localhost:4200
Student Management
ID
Name
Gender
Date of Birth
Course
Marks
Percentage
Fees
1 John Smith Male NOVEMBER 12, 1997 MBA 520 86.67% ₹1,899.00
2 Mark Vought Male OCTOBER 6, 1998 B.TECH 420 70.00% ₹2,899.00
3 Sarah King Female SEPTEMBER 22, 1996 B.TECH 540 90.00% ₹2,899.00
4 Merry Jane Female JUNE 12, 1995 MBA 380 63.33% ₹1,899.00
5 Steve Smith Male DECEMBER 21, 1999 M.SC 430 71.67% ₹799.00
6 Jonas Weber Male JUNE 18, 1997 M.SC 320 53.33% ₹799.00
7 TEST Female AUGUST 15, 2023 B.SC 320 53.33% ₹1,299.00
Elements
Console
Sources
Network
Performance
Memory
Application
Security
Lighthouse
Recorder
Performance insights
top
Filter
Default levels
1 Issue: 1
FILTER PIPE CALLED!
filter.pipe.ts:9
PERCENTAGE PIPE CALLED!
percentage.pipe.ts:9
Warning: the currency pipe has been changed in Angular v5. The symbolDisplay option (third parameter) is now a string instead of a boolean. The accepted values are "code", "symbol" or "symbol-narrow".
admin.component.html:121

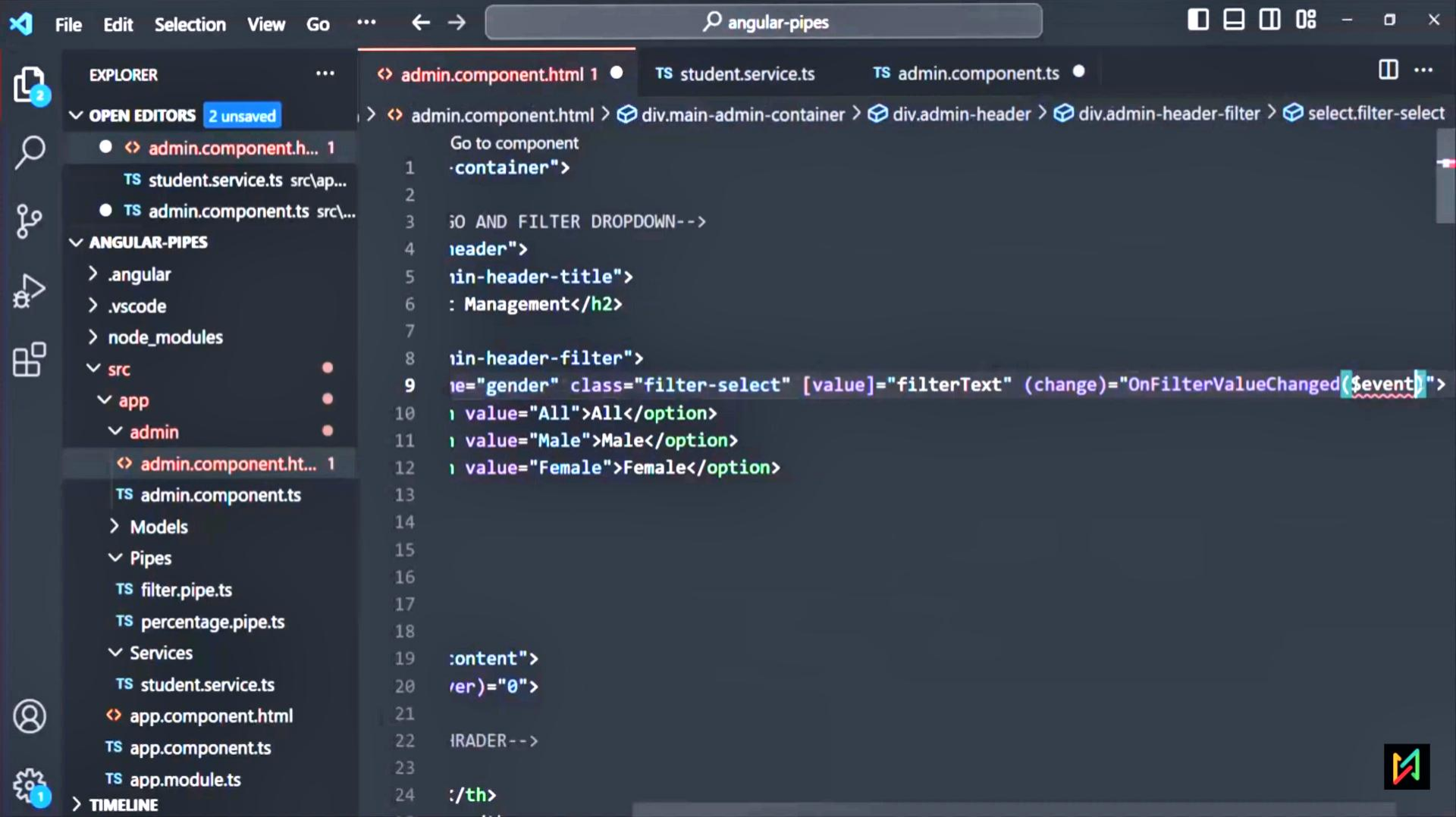
What is an impure pipe?

A pipe is impure pipe if it is not pure. An impure pipe gets called for each change detection cycle and it is performance intensive.


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
student.service.ts
percentage.pipe.ts
filter.pipe.ts
OPEN EDITORS 1 unsaved
admin.component.html...
student.service.ts src\ap...
percentage.pipe.ts src\a...
filter.pipe.ts src\app\Pipes
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
student.service.ts
app.component.html
app.component.ts
TIMELINE
src > app > Pipes > filter.pipe.ts > FilterPipe
1 import { Pipe, PipeTransform } from "@angular/core";
2 import { Student } from "../Models/Student";
3
4 @Pipe({
5 name: 'filter',
6 pure: false
7 })
8 export class FilterPipe implements PipeTransform{
9 transform(list: Student[], filterBy: string) {
10 console.log('FILTER PIPE CALLED!');
11 if(filterBy.toLowerCase() === 'all' || filterBy === '' || list.length === 0){
12 return list;
13 }else{
14 return list.filter((std) => {
15 return std.gender.toLowerCase() === filterBy.toLowerCase();
16 })
17 }
18 }
19 }

Adding Filter Functionality without Pipes


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
student.service.ts
percentage.pipe.ts
filter.pipe.ts
OPEN EDITORS 2 unsaved
admin.component.html...
student.service.ts src\ap...
percentage.pipe.ts src\a...
filter.pipe.ts src\app\Pipes
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
student.service.ts
app.component.html
app.component.ts
TIMELINE
src > app > Services > student.service.ts > StudentService > filterStudentByGender
11];//0x12345
12
13 totalMarks: number = 600;
14
15 CreateStudent(name, gender, dob, course, marks, fee){
16 let id = this.students.length + 1;
17 let student = new Student(id, name, gender, dob, course, marks, fee);
18 this.students.push(student);
19
20 // let studentCopy = [...this.students];//0x34567
21 // studentCopy.push(student);
22 // this.students = studentCopy;
23 }
24
25 filterStudentByGender(filterBy: string){
26 if(filterBy.toLowerCase() === 'all' || filterBy === '' || this.students.length === 0){
27 return this.students;
28 }else{
29 return this.students.filter((std) => {
30 return std.gender.toLowerCase() === filterBy.toLowerCase();
31 })
32 }
33 }
34 }


File Edit Selection View Go
angular-pipes
EXPLORER
OPEN EDITORS 2 unsaved
admin.component.h... 1
student.service.ts src\ap...
admin.component.ts src\...
ANGULAR-PIPES
.angular
.vscode
node_modules
src
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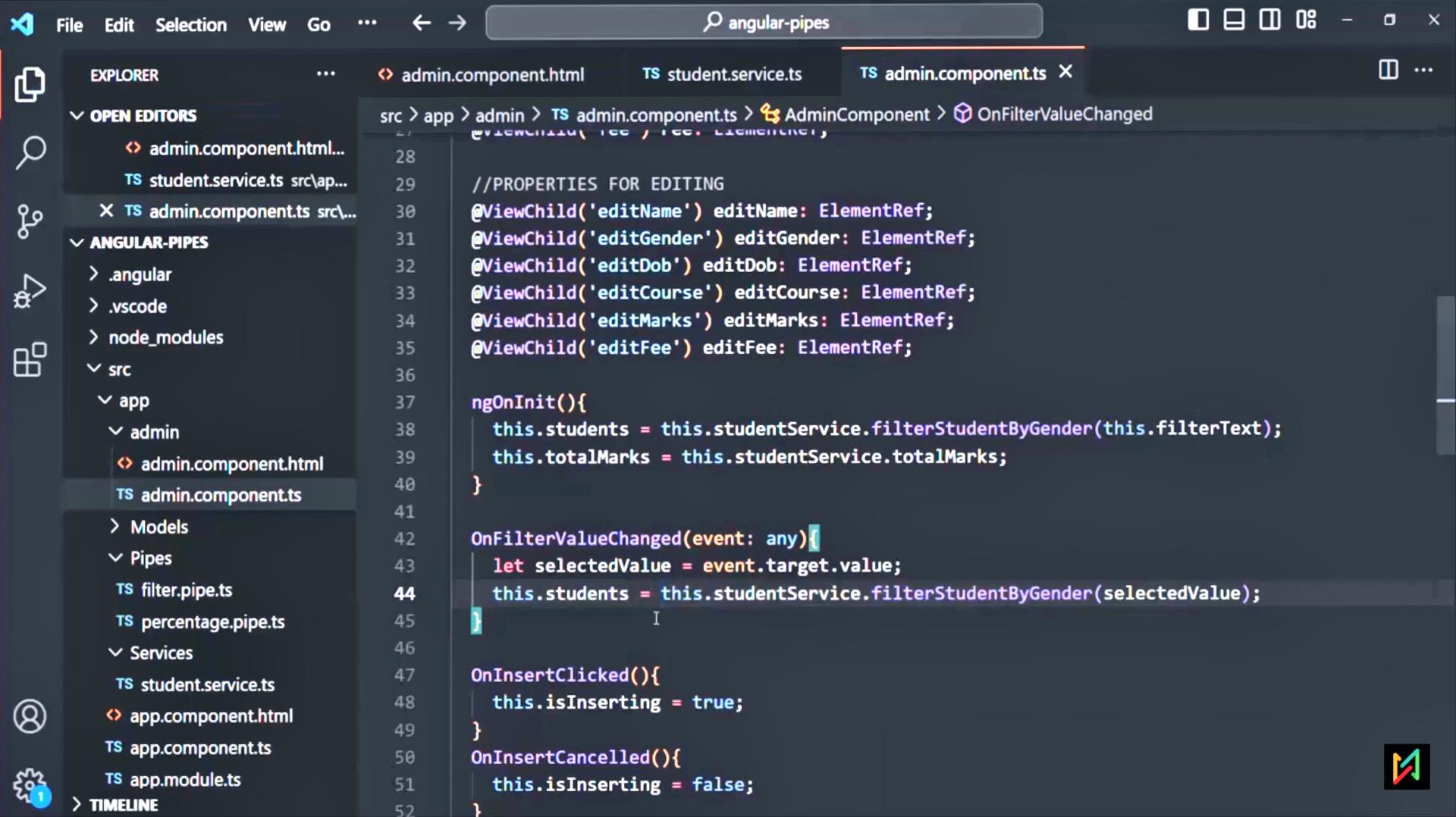
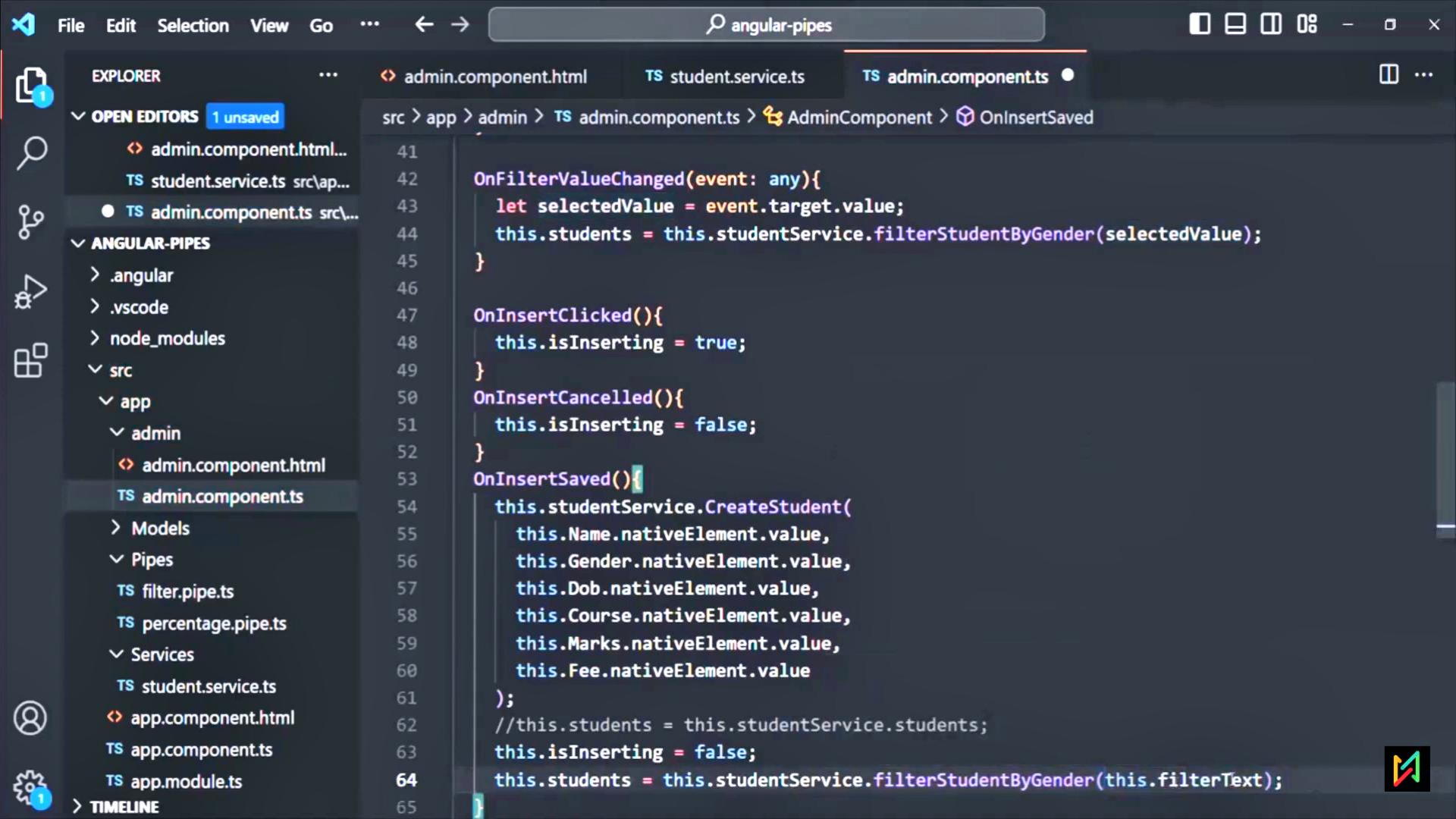
admin.component.ht... 1
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
student.service.ts
app.component.html
app.component.ts
app.module.ts
TIMELINE
admin.component.html 1
student.service.ts
admin.component.ts
admin.component.html > div.main-admin-container > div.admin-header > div.admin-header-filter > select.filter-select
Go to component
container">
GO AND FILTER DROPDOWN-->
eader">
in-header-title">
: Management</h2>
in-header-filter">
e="gender" class="filter-select" [value]="filterText" (change)="OnFilterValueChanged($event)">
value="All">All</option>
value="Male">Male</option>
value="Female">Female</option>
ontent">
er)="0">
HRADER-->
:/th>


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
student.service.ts
admin.component.ts
src > app > admin > admin.component.ts > AdminComponent > OnFilterValueChanged
OPEN EDITORS
admin.component.html...
student.service.ts src\ap...
admin.component.ts src\...
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
student.service.ts
app.component.html
app.component.ts
app.module.ts
TIMELINE
28
29 //PROPERTIES FOR EDITING
30 @ViewChild('editName') editName: ElementRef;
31 @ViewChild('editGender') editGender: ElementRef;
32 @ViewChild('editDob') editDob: ElementRef;
33 @ViewChild('editCourse') editCourse: ElementRef;
34 @ViewChild('editMarks') editMarks: ElementRef;
35 @ViewChild('editFee') editFee: ElementRef;
36
37 ngOnInit(){
38 this.students = this.studentService.filterStudentByGender(this.filterText);
39 this.totalMarks = this.studentService.totalMarks;
40 }
41
42 OnFilterValueChanged(event: any){
43 let selectedValue = event.target.value;
44 this.students = this.studentService.filterStudentByGender(selectedValue);
45 }
46
47 OnInsertClicked(){
48 this.isInserting = true;
49 }
50 OnInsertCancelled(){
51 this.isInserting = false;
52 }


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
student.service.ts
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OPEN EDITORS 1 unsaved
admin.component.html...
student.service.ts src\ap...
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ANGULAR-PIPES
.angular
.vscode
node_modules
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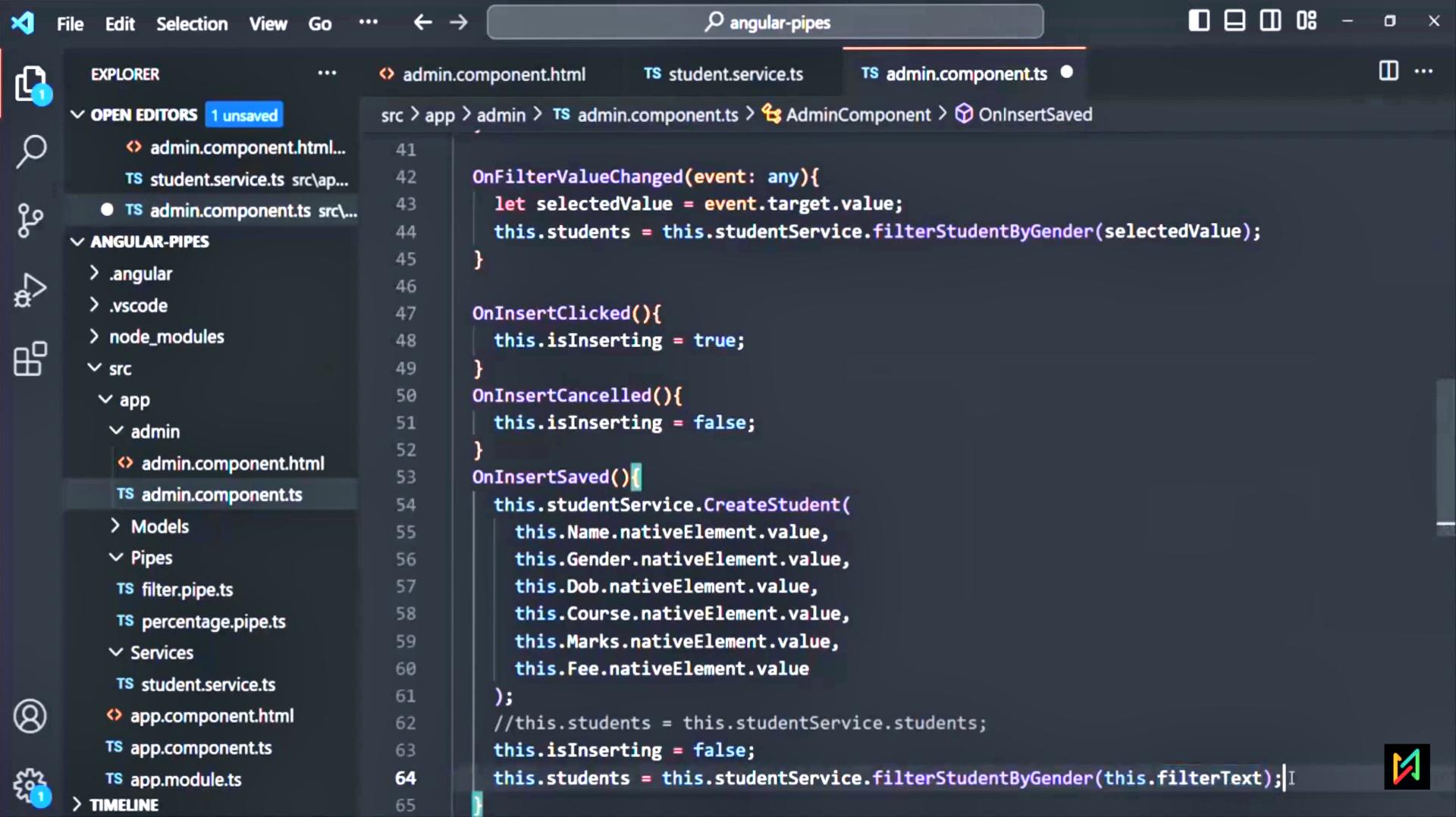
admin.component.html
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
student.service.ts
app.component.html
app.component.ts
app.module.ts
TIMELINE
src > app > admin > admin.component.ts > AdminComponent > OnInsertSaved
41
42 OnFilterValueChanged(event: any){
43 let selectedValue = event.target.value;
44 this.students = this.studentService.filterStudentByGender(selectedValue);
45 }
46
47 OnInsertClicked(){
48 this.isInserting = true;
49 }
50 OnInsertCancelled(){
51 this.isInserting = false;
52 }
53 OnInsertSaved(){
54 this.studentService.CreateStudent(
55 this.Name.nativeElement.value,
56 this.Gender.nativeElement.value,
57 this.Dob.nativeElement.value,
58 this.Course.nativeElement.value,
59 this.Marks.nativeElement.value,
60 this.Fee.nativeElement.value
61);
62 //this.students = this.studentService.students;
63 this.isInserting = false;
64 this.students = this.studentService.filterStudentByGender(this.filterText);
65 }


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
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admin.component.ts
src > app > admin > admin.component.ts > AdminComponent > OnInsertSaved
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admin.component.ts src\...
ANGULAR-PIPES
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.vscode
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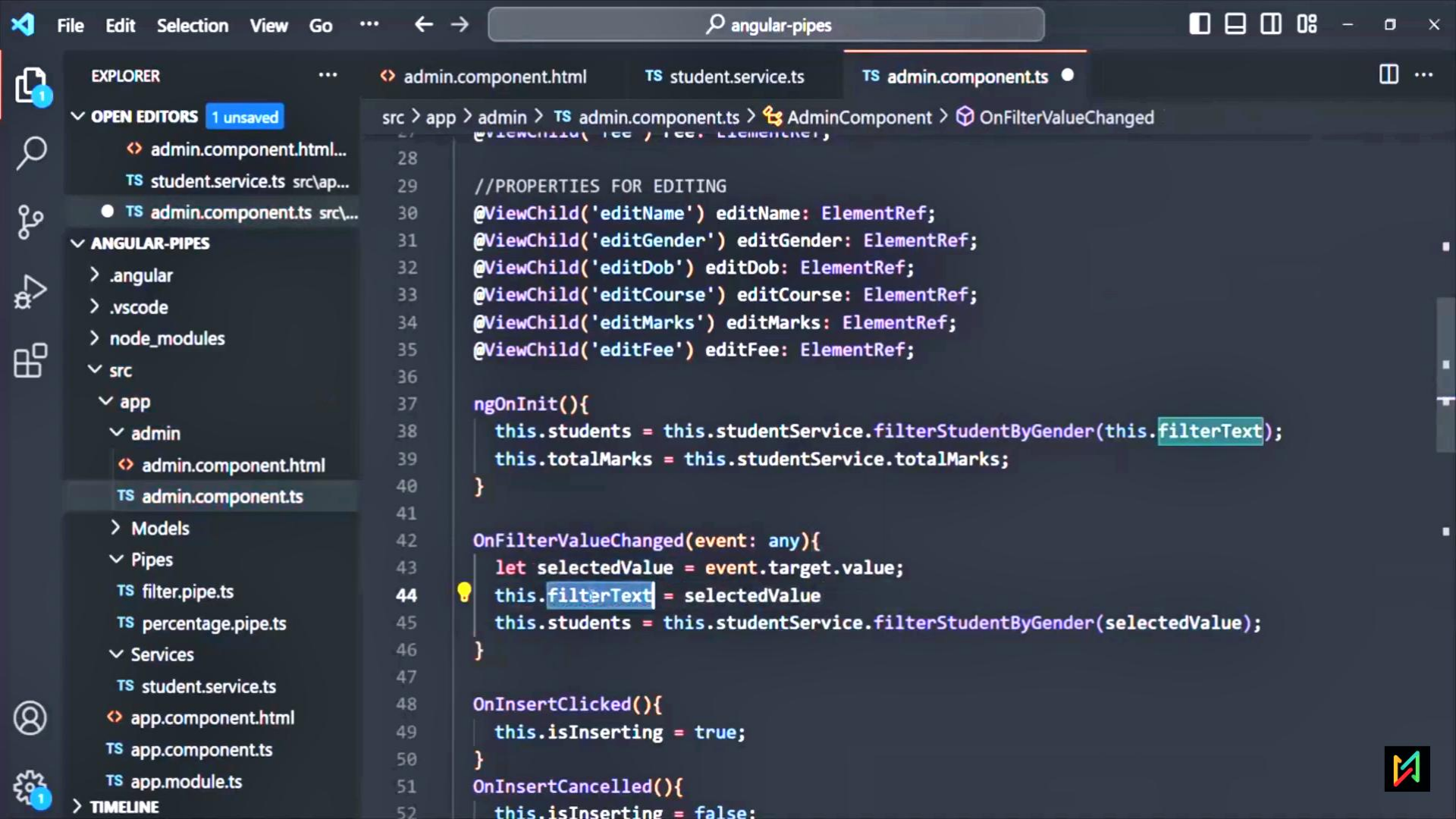
admin.component.html
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
student.service.ts
app.component.html
app.component.ts
app.module.ts
TIMELINE
41
42 OnFilterValueChanged(event: any){
43 let selectedValue = event.target.value;
44 this.students = this.studentService.filterStudentByGender(selectedValue);
45 }
46
47 OnInsertClicked(){
48 this.isInserting = true;
49 }
50 OnInsertCancelled(){
51 this.isInserting = false;
52 }
53 OnInsertSaved(){
54 this.studentService.CreateStudent(
55 this.Name.nativeElement.value,
56 this.Gender.nativeElement.value,
57 this.Dob.nativeElement.value,
58 this.Course.nativeElement.value,
59 this.Marks.nativeElement.value,
60 this.Fee.nativeElement.value
61);
62 //this.students = this.studentService.students;
63 this.isInserting = false;
64 this.students = this.studentService.filterStudentByGender(this.filterText);
65 }


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
student.service.ts
admin.component.ts
OPEN EDITORS 1 unsaved
admin.component.html...
student.service.ts src\ap...
admin.component.ts src\...
ANGULAR-PIPES
.angular
.vscode
node_modules
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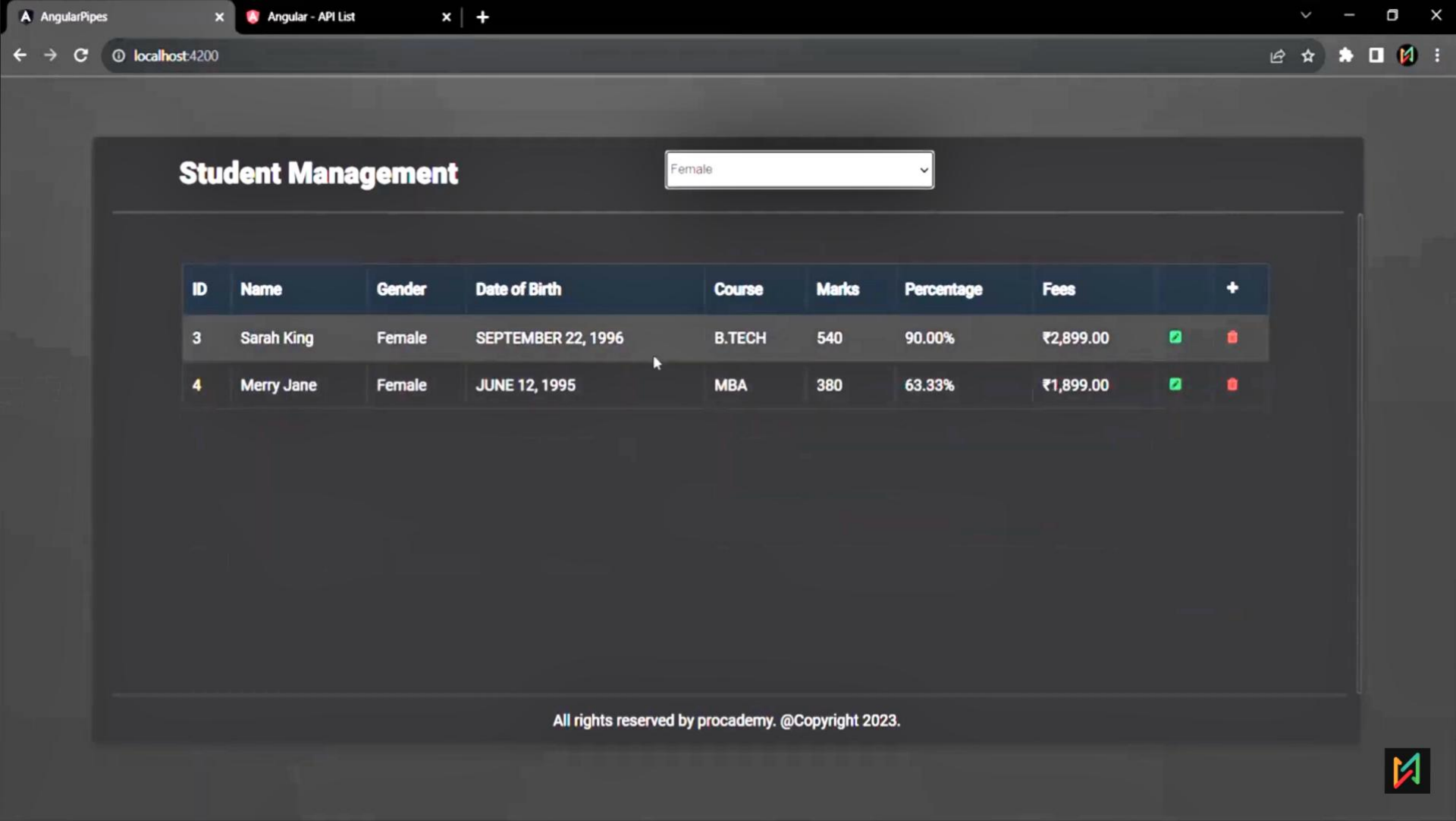
admin.component.html
admin.component.ts
Models
Pipes
filter.pipe.ts
percentage.pipe.ts
Services
student.service.ts
app.component.html
app.component.ts
app.module.ts
TIMELINE
src > app > admin > admin.component.ts > AdminComponent > OnFilterValueChanged
28
29 //PROPERTIES FOR EDITING
30 @ViewChild('editName') editName: ElementRef;
31 @ViewChild('editGender') editGender: ElementRef;
32 @ViewChild('editDob') editDob: ElementRef;
33 @ViewChild('editCourse') editCourse: ElementRef;
34 @ViewChild('editMarks') editMarks: ElementRef;
35 @ViewChild('editFee') editFee: ElementRef;
36
37 ngOnInit(){
38 this.students = this.studentService.filterStudentByGender(this.filterText);
39 this.totalMarks = this.studentService.totalMarks;
40 }
41
42 OnFilterValueChanged(event: any){
43 let selectedValue = event.target.value;
44 this.filterText = selectedValue
45 this.students = this.studentService.filterStudentByGender(selectedValue);
46 }
47
48 OnInsertClicked(){
49 this.isInserting = true;
50 }
51 OnInsertCancelled(){
52 this.isInserting = false;


AngularPipes
Angular - API List
localhost:4200
Student Management
Female
ID
Name
Gender
Date of Birth
Course
Marks
Percentage
Fees
3
Sarah King
Female
SEPTEMBER 22, 1996
B.TECH
540
90.00%
₹2,899.00
4
Merry Jane
Female
JUNE 12, 1995
MBA
380
63.33%
₹1,899.00
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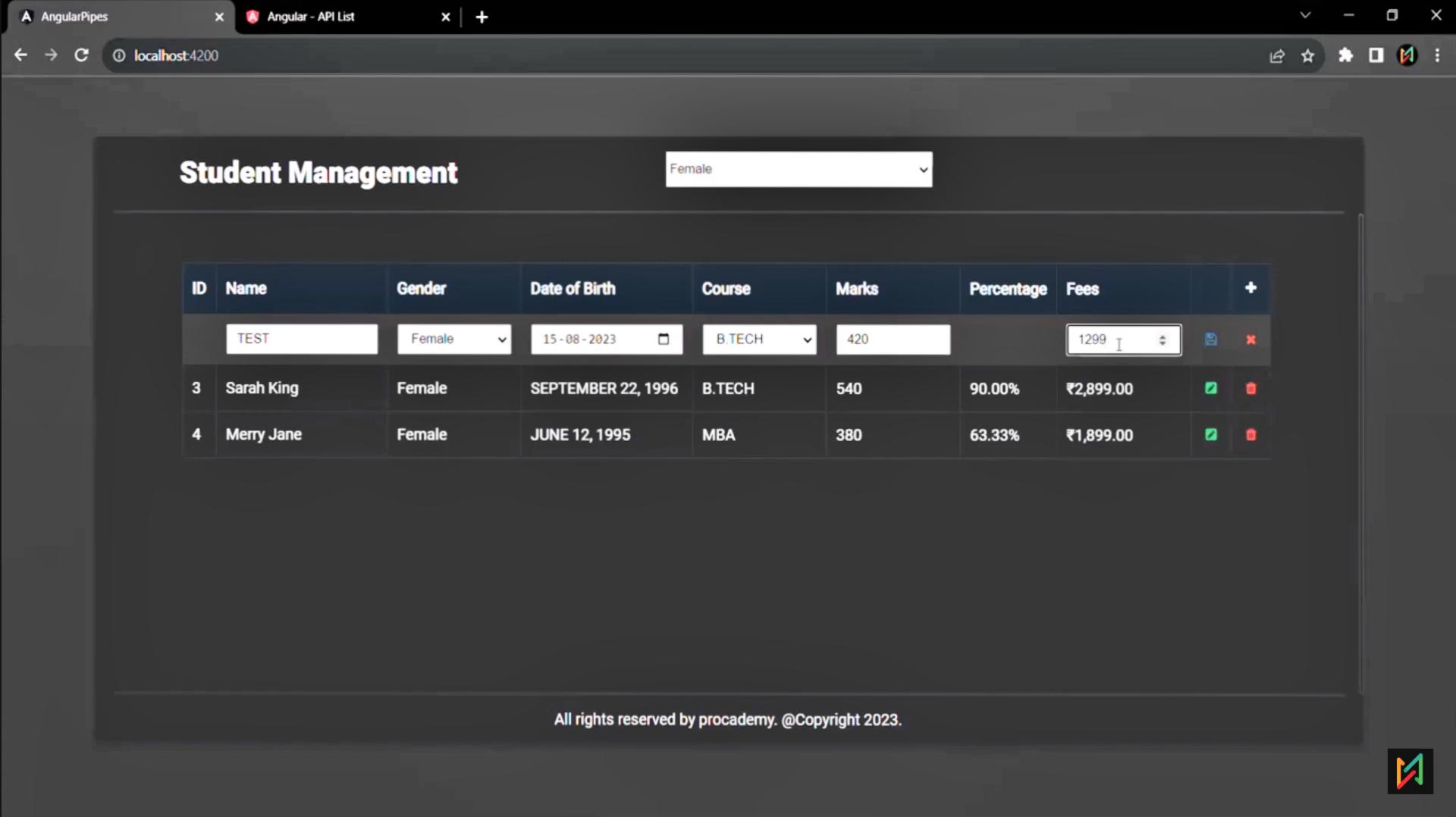
AngularPipes
Angular - API List
localhost:4200
Student Management
Female
ID
Name
Gender
Date of Birth
Course
Marks
Percentage
Fees
TEST
Female
15-08-2023
B.TECH
420
1299
3
Sarah King
Female
SEPTEMBER 22, 1996
B.TECH
540
90.00%
₹2,899.00
4
Merry Jane
Female
JUNE 12, 1995
MBA
380
63.33%
₹1,899.00
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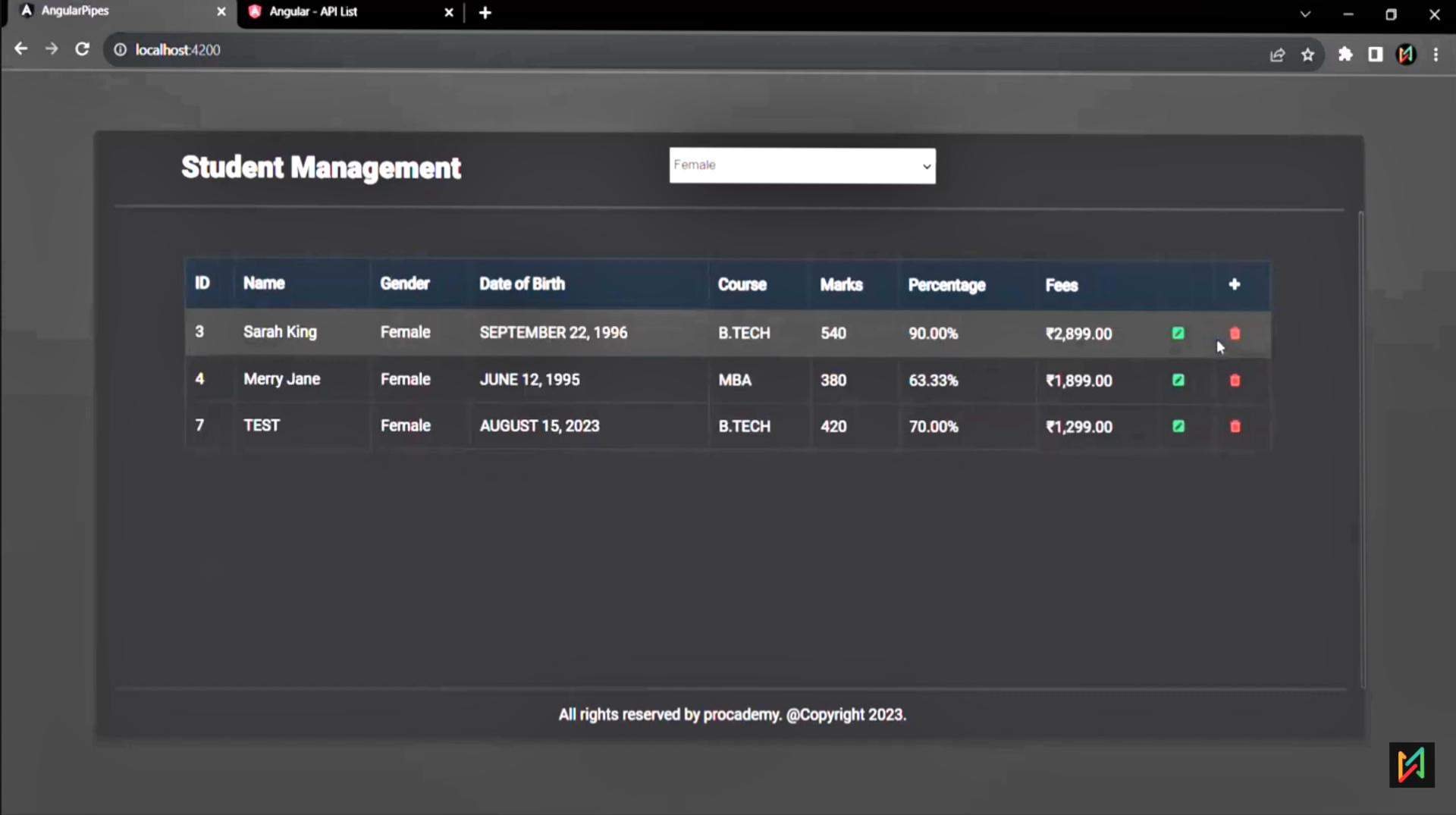
AngularPipes
Angular - API List
localhost:4200
Student Management
Female
ID
Name
Gender
Date of Birth
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Marks
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Merry Jane
Female
JUNE 12, 1995
MBA
380
63.33%
₹1,899.00
7
TEST
Female
AUGUST 15, 2023
B.TECH
420
70.00%
₹1,299.00
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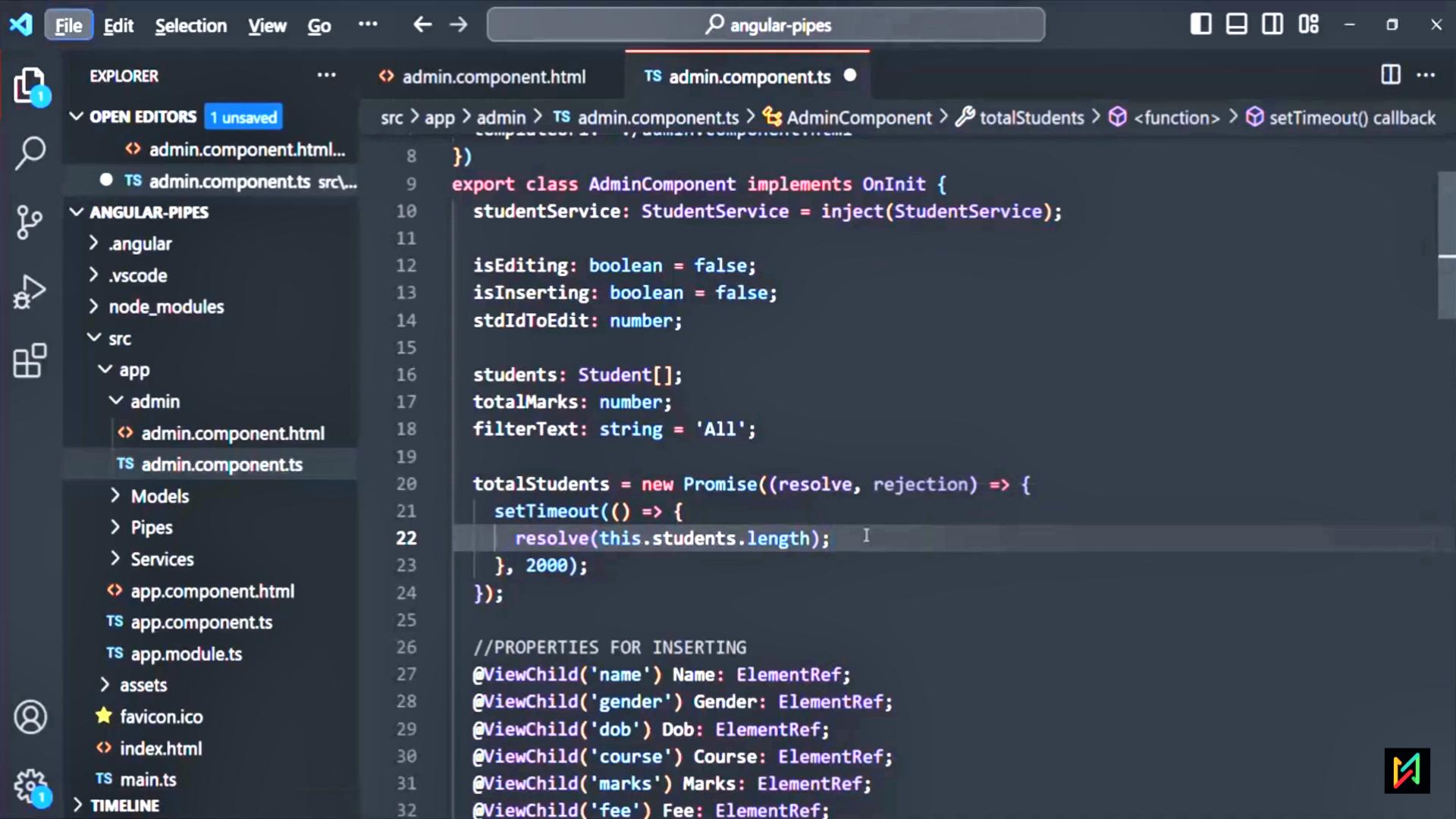
async pipe

The async pipe allows us to handle asynchronous data. It allows us to subscribe to an observable or promise from the view template and returns the value emitted.


File Edit Selection View Go
angular-pipes
EXPLORER
admin.component.html
admin.component.ts
OPEN EDITORS 1 unsaved
admin.component.html...
admin.component.ts src\...
ANGULAR-PIPES
.angular
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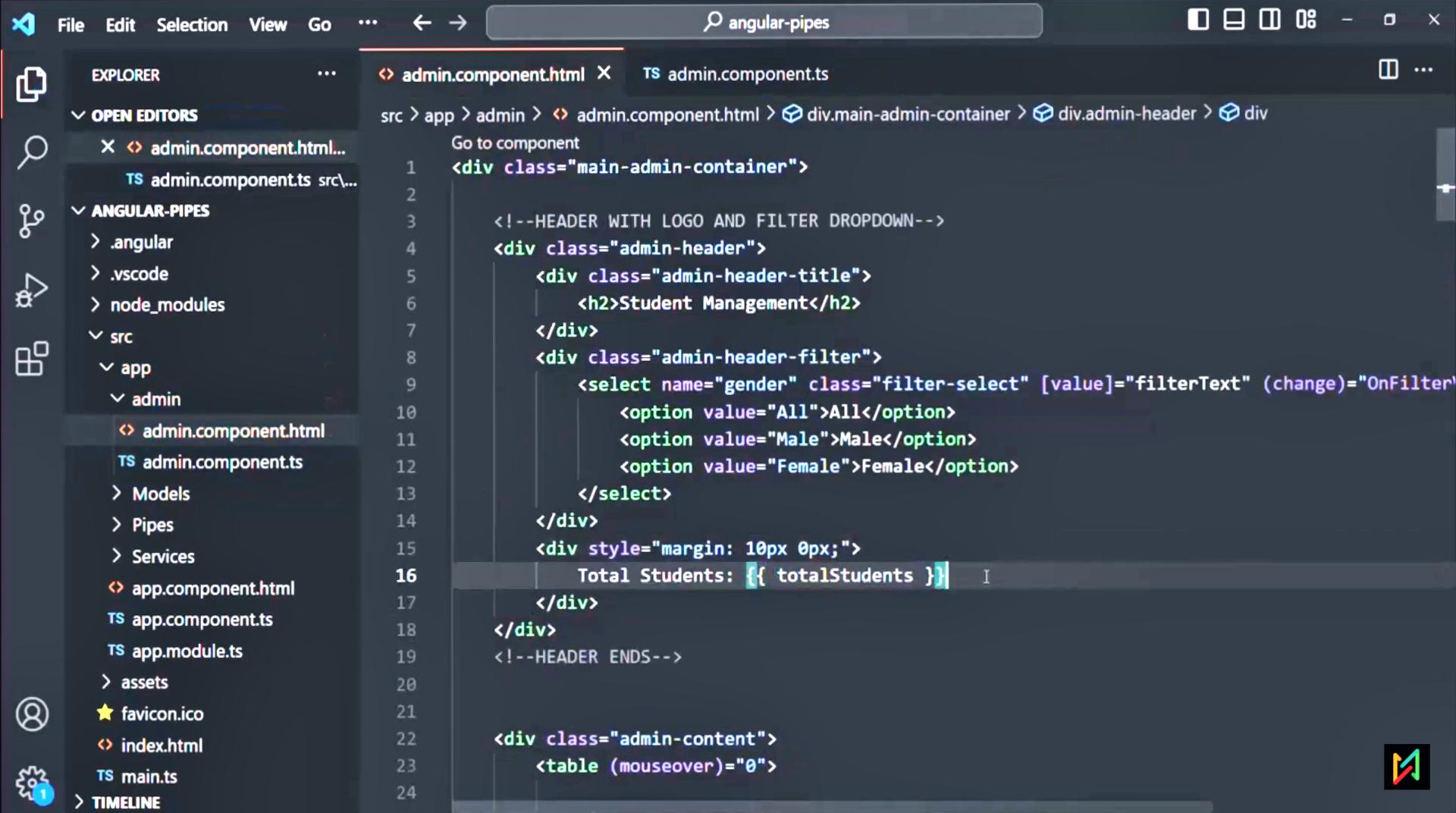
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
index.html
main.ts
TIMELINE
src > app > admin > admin.component.ts > AdminComponent > totalStudents > <function> > setTimeout() callback
8 })
9 export class AdminComponent implements OnInit {
10 studentService: StudentService = inject(StudentService);
11
12 isEditing: boolean = false;
13 isInserting: boolean = false;
14 stdIdToEdit: number;
15
16 students: Student[];
17 totalMarks: number;
18 filterText: string = 'All';
19
20 totalStudents = new Promise((resolve, rejection) => {
21 setTimeout(() => {
22 resolve(this.students.length);
23 }, 2000);
24 });
25
26 //PROPERTIES FOR INSERTING
27 @ViewChild('name') Name: ElementRef;
28 @ViewChild('gender') Gender: ElementRef;
29 @ViewChild('dob') Dob: ElementRef;
30 @ViewChild('course') Course: ElementRef;
31 @ViewChild('marks') Marks: ElementRef;
32 @ViewChild('fee') Fee: ElementRef;


File Edit Selection View Go
angular-pipes
EXPLORER
OPEN EDITORS
admin.component.html...
admin.component.ts src\...
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
index.html
main.ts
TIMELINE
admin.component.html
admin.component.ts
src > app > admin > admin.component.html > div.main-admin-container > div.admin-header > div
Go to component

<!--HEADER WITH LOGO AND FILTER DROPDOWN-->

<h2>Student Management</h2>

<select name="gender" class="filter-select" [value]="filterText" (change)="OnFilter
<option value="All">All</option>
<option value="Male">Male</option>
<option value="Female">Female</option>
</select>

Total Students: {{ totalStudents }}

<!--HEADER ENDS-->

<table (mouseover)="0">
1
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3
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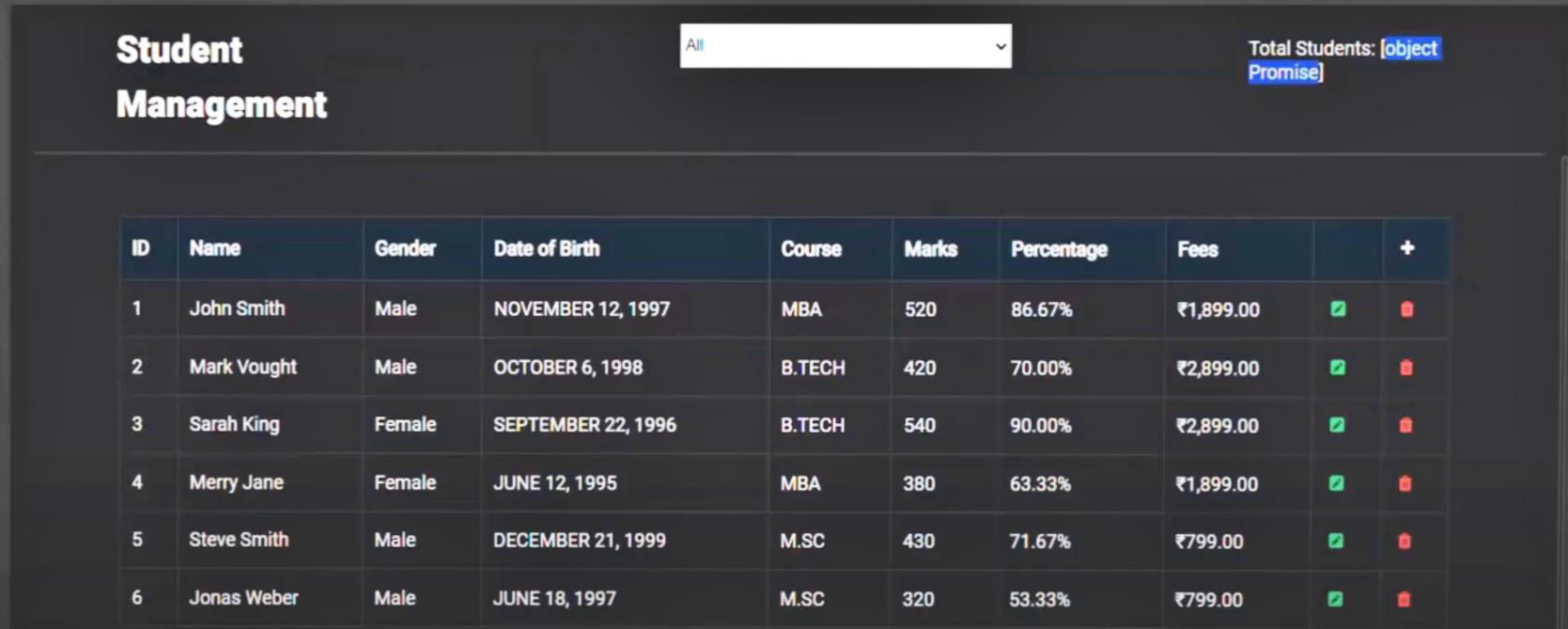
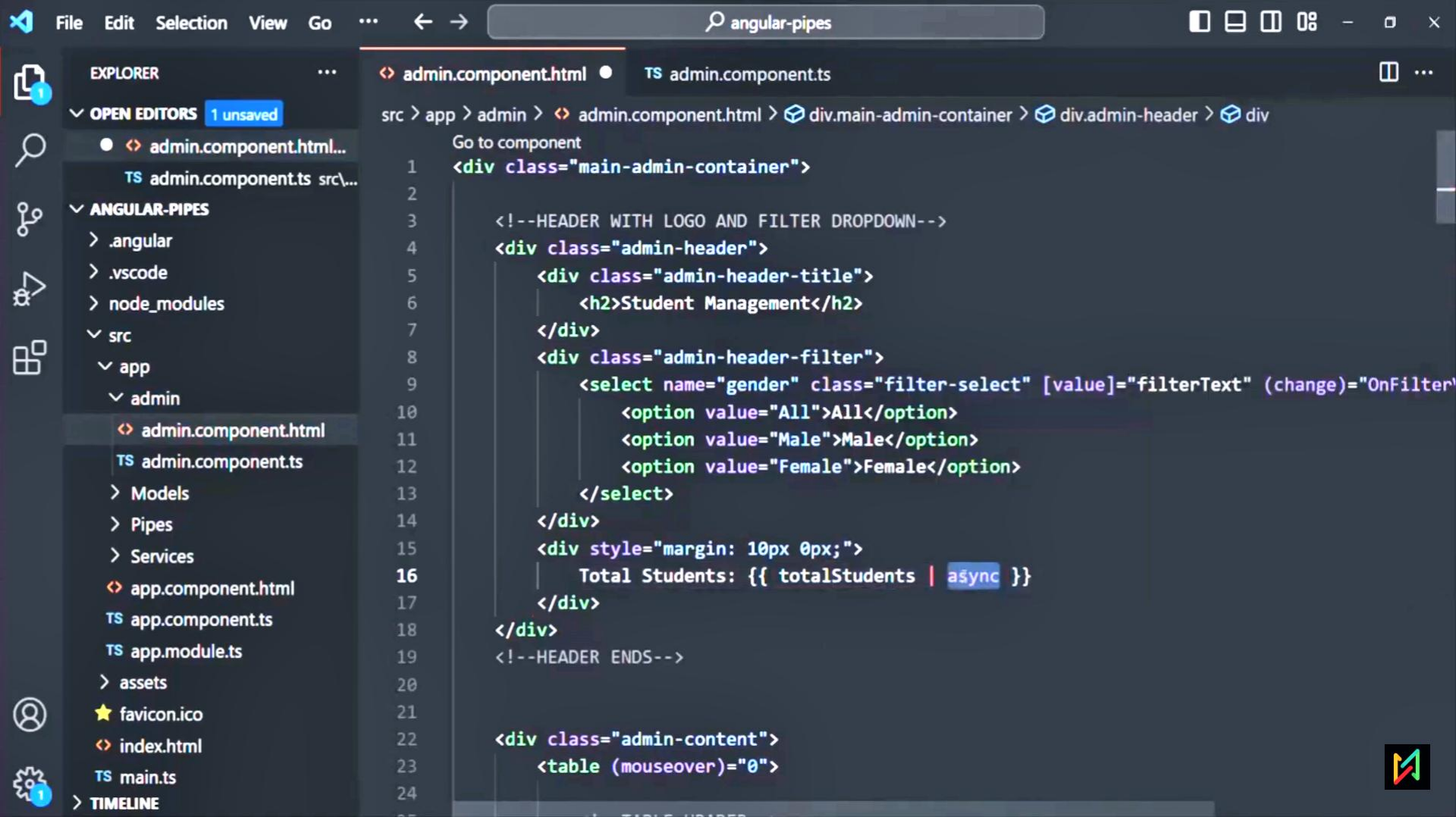
#107 Async Pipe in Angular | Understanding Angular Pipes | A Complete Angular Course



6:48 / 7:42


File Edit Selection View Go
angular-pipes
EXPLORER
OPEN EDITORS 1 unsaved
admin.component.html...
admin.component.ts src\...
ANGULAR-PIPES
.angular
.vscode
node_modules
src
app
admin
admin.component.html
admin.component.ts
Models
Pipes
Services
app.component.html
app.component.ts
app.module.ts
assets
favicon.ico
index.html
main.ts
TIMELINE
admin.component.html
admin.component.ts
src > app > admin > admin.component.html > div.main-admin-container > div.admin-header > div
Go to component

<!--HEADER WITH LOGO AND FILTER DROPDOWN-->

<h2>Student Management</h2>

<select name="gender" class="filter-select" [value]="filterText" (change)="OnFilter
<option value="All">All</option>
<option value="Male">Male</option>
<option value="Female">Female</option>
</select>

Total Students: {{ totalStudents | async }}

<!--HEADER ENDS-->

<table (mouseover)="0">
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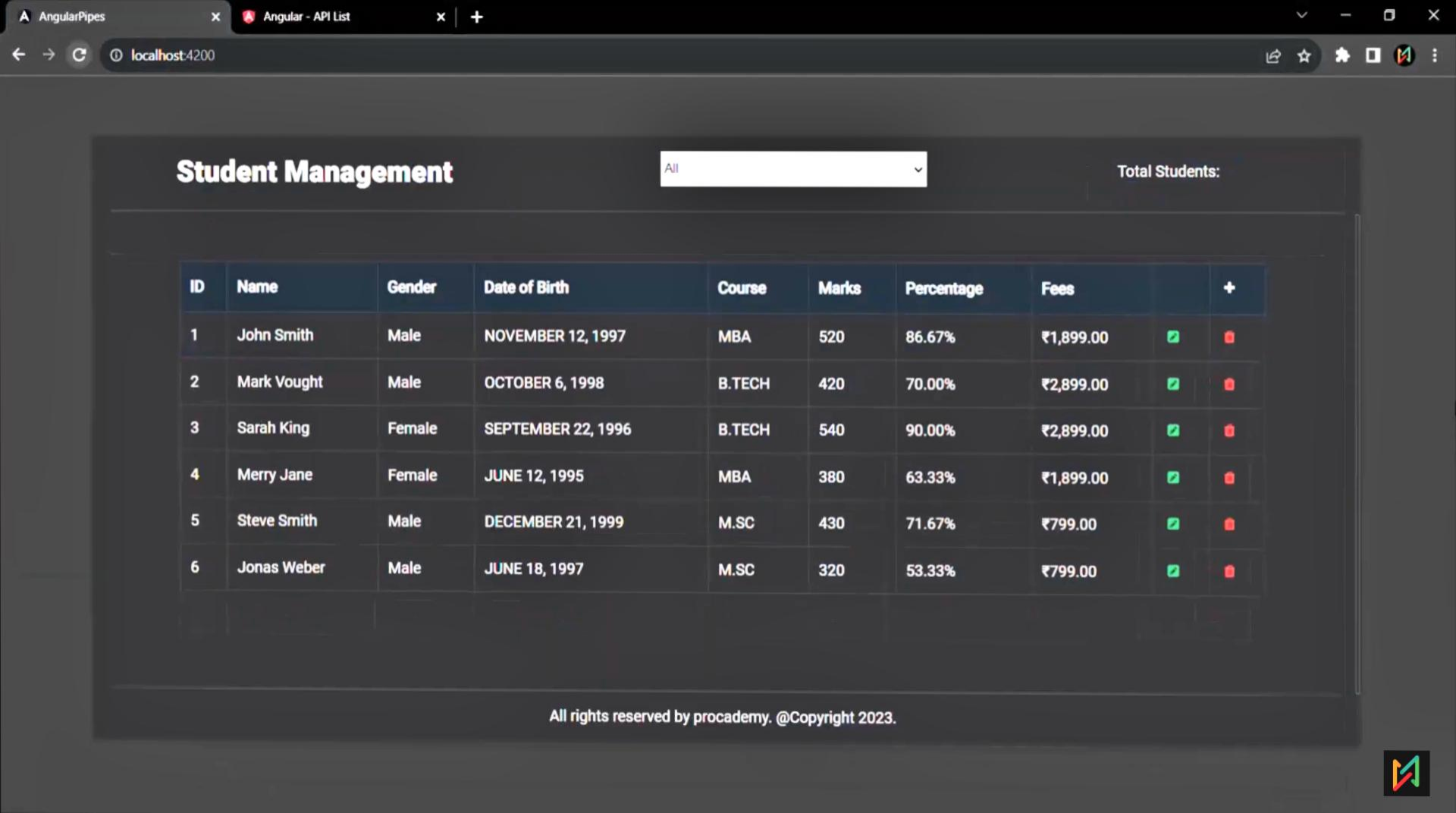
AngularPipes
Angular - API List
localhost:4200
Student Management
All
Total Students:
ID
Name
Gender
Date of Birth
Course
Marks
Percentage
Fees
1
John Smith
Male
NOVEMBER 12, 1997
MBA
520
86.67%
₹1,899.00
2
Mark Vought
Male
OCTOBER 6, 1998
B.TECH
420
70.00%
₹2,899.00
3
Sarah King
Female
SEPTEMBER 22, 1996
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380
63.33%
₹1,899.00
5
Steve Smith
Male
DECEMBER 21, 1999
M.SC
430
71.67%
₹799.00
6
Jonas Weber
Male
JUNE 18, 1997
M.SC
320
53.33%
₹799.00
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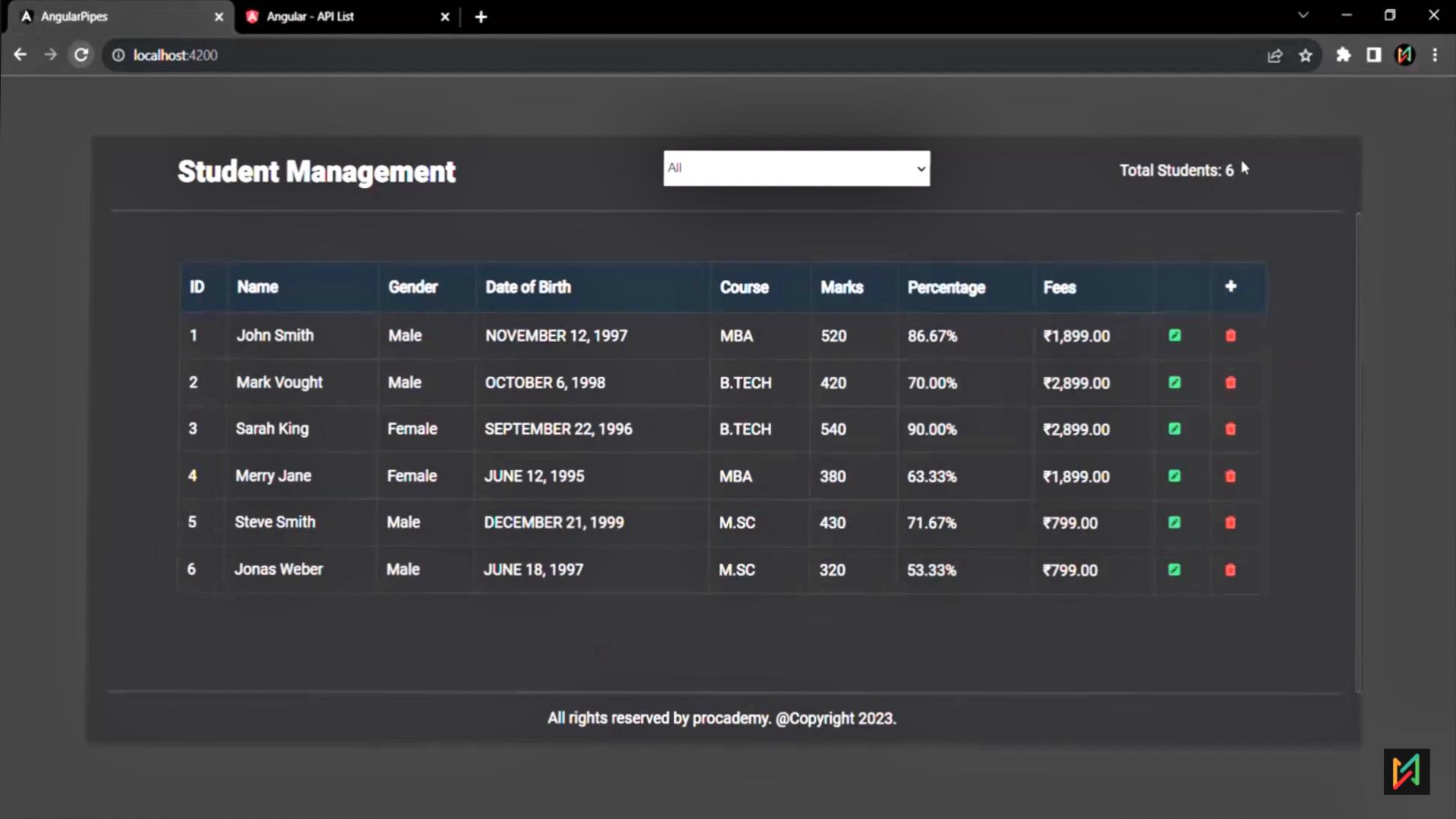
AngularPipes
Angular - API List
localhost:4200
Student Management
All
Total Students: 6
ID
Name
Gender
Date of Birth
Course
Marks
Percentage
Fees
+
1 John Smith Male NOVEMBER 12, 1997 MBA 520 86.67% ₹1,899.00
2 Mark Vought Male OCTOBER 6, 1998 B.TECH 420 70.00% ₹2,899.00
3 Sarah King Female SEPTEMBER 22, 1996 B.TECH 540 90.00% ₹2,899.00
4 Merry Jane Female JUNE 12, 1995 MBA 380 63.33% ₹1,899.00
5 Steve Smith Male DECEMBER 21, 1999 M.SC 430 71.67% ₹799.00
6 Jonas Weber Male JUNE 18, 1997 M.SC 320 53.33% ₹799.00
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